Introduction to Big Data

DATA MANAGEMENT

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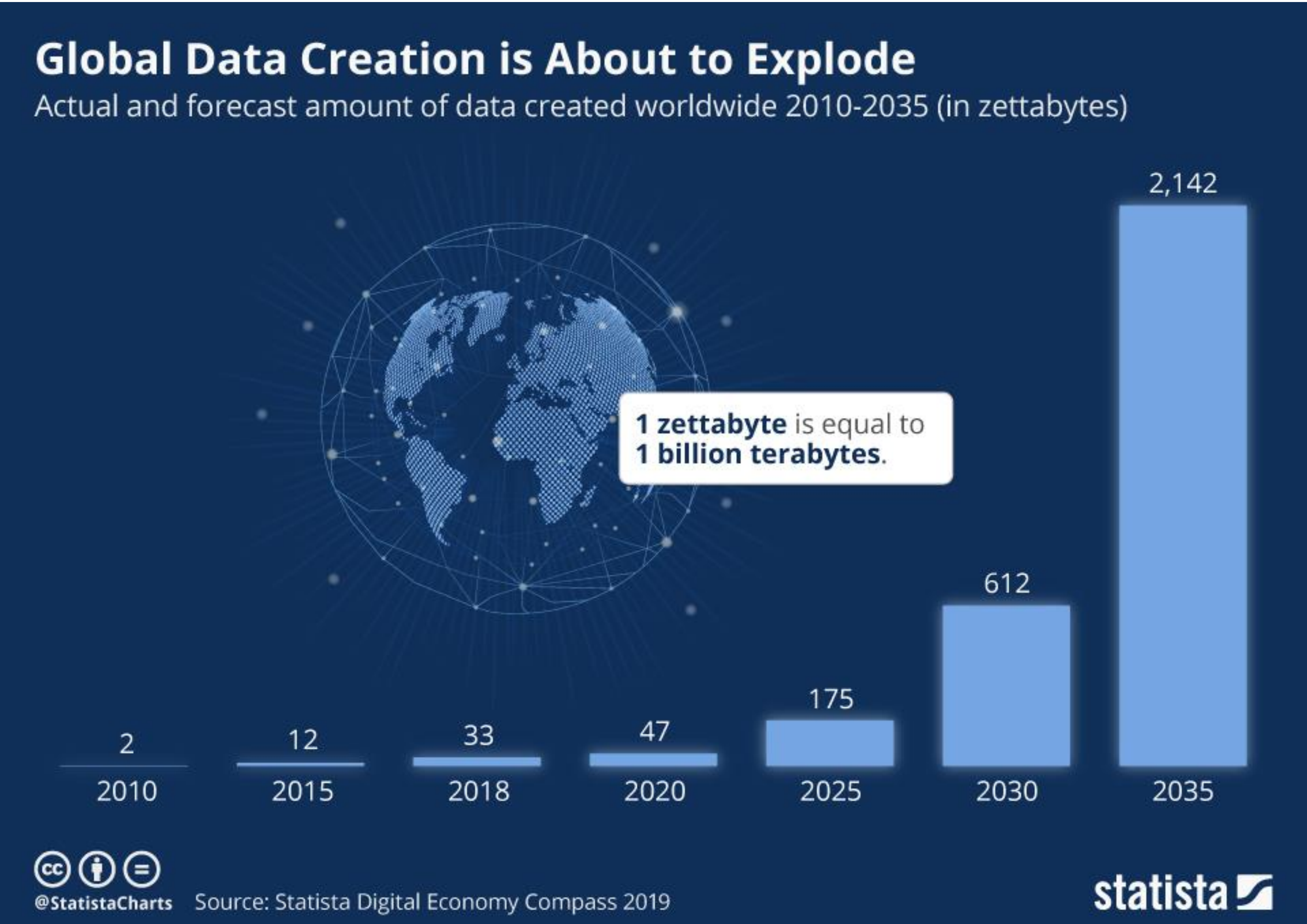
Content outline

- Data in the world
- Data repository techniques
- NoSQL databases


Global Data Creation is About to Explode
Actual and forecast amount of data created worldwide 2010-2035 (in zettabytes)
1 zettabyte is equal to 1 billion terabytes.
2
12
33
47
175
612
2,142
2010
2015
2018
2020
2025
2030
2035
@StatistaCharts
Source: Statista Digital Economy Compass 2019
statista

A DAY IN DATA

The exponential growth of data is undisputed, but the numbers behind this explosion – fuelled by internet of things and the use of connected devcies – are hard to comprehend, particularly when looked at in the context of one day

500m

tweets are sent every day

Twitter

4PB

of data created by Facebook, including

350m photos

100m hours of video watch time

Facebook Research

DEMYSTIFIYING DATA UNITS

From the more familiar 'bit' or 'megabyte', larger units of measurement are more frequently being used to explain the masses of data

Unit		Value	Size
b	bit	0 or 1	1/8 of a byte
B	byte	8 bits	1 byte
KB	kilobyte	1,000 bytes	1,000 bytes
MB	megabyte	$1{,}000^2$ bytes	1,000,000 bytes
GB	gigabyte	$1{,}000^3$ bytes	1,000,000,000 bytes
TB	terabyte	$1{,}000^4$ bytes	1,000,000,000,000 bytes
PB	petabyte	$1{,}000^5$ bytes	1,000,000,000,000,000 bytes
EB	exabyte	$1{,}000^6$ bytes	1,000,000,000,000,000,000 bytes
ZB	zettabyte	$1{,}000^7$ bytes	1,000,000,000,000,000,000,000 bytes
YB	yottabyte	$1{,}000^8$ bytes	1,000,000,000,000,000,000,000,000 bytes

*A lowercase "b" is used as an abbreviation for bits, while an uppercase "B" represents bytes.

463EB

of data will be created every day by 2025

IDC

95m

photos and videos are shared on Instagram

Instagram Business

294bn

billion emails are sent

Radicati Group

320bn

emails to be sent each day by 2021

306bn

emails to be sent each day by 2020

65bn

messages sent over WhatsApp and two billion minutes of voice and video calls made

Facebook

3.9bn

people use emails

4TB

of data produced by a connected car

Intel

Searches made a day – 5bn

Searches made a day from Google – 3.5bn

Smart Insights

28PB

to be generated from wearable devices by 2020

Statista

ACCUMULATED DIGITAL UNIVERSE OF DATA

4.4ZB – 2013

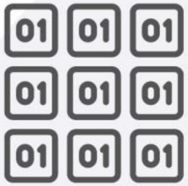
44ZB – 2020

PwC

RACONTEUR

From **80 to 90 percent of data** generated and collected by organizations is **unstructured**.

Structured data



Characteristics

Predefined data models
Easy to search
Text-based
Shows what's happening

Resides in

Relational databases
Data warehouses

Stored in

Rows and columns

Examples

Dates, phone numbers, social security numbers, customer names, transaction info

Unstructured data

Characteristics

No predefined data models
Difficult to search
Text, pdf, images, video
Shows the why

Resides in

Applications
Data warehouses and lakes

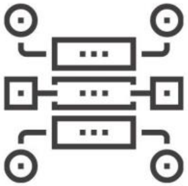
Stored in

Various forms

Examples

Documents, emails and messages, conversation transcripts, image files, open-ended survey answers

Semi-structured data



Characteristics

Loosely organized
Meta-level structure that can contain unstructured data
HTML, XML, JSON

Resides in

Relational databases
Tagged-text format

Stored in

Abstracts & figures

Examples

Server logs, tweets organized by hashtags, emails sorting by folders (inbox; sent; draft)

LEVITY

- Data in the world has evolved with new characteristics.

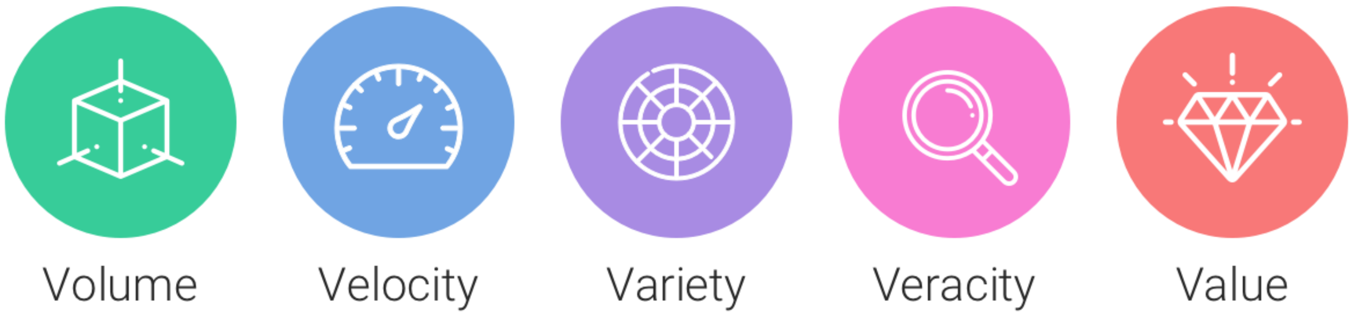


Image credit: Mailjet

- New technologies for handling data efficiently are needed.

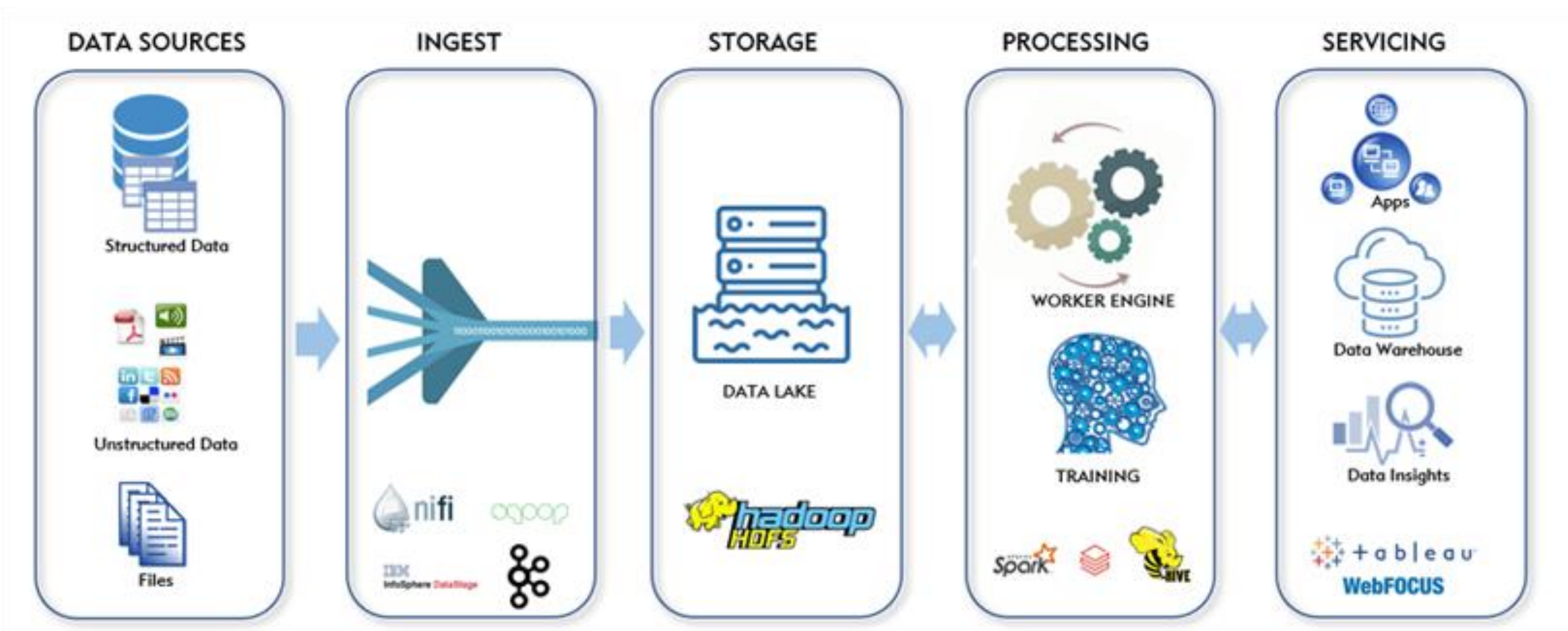


Image credit: Narwal Inc.

Data repository techniques

The data in an enterprise may come from different sources and in various formats.

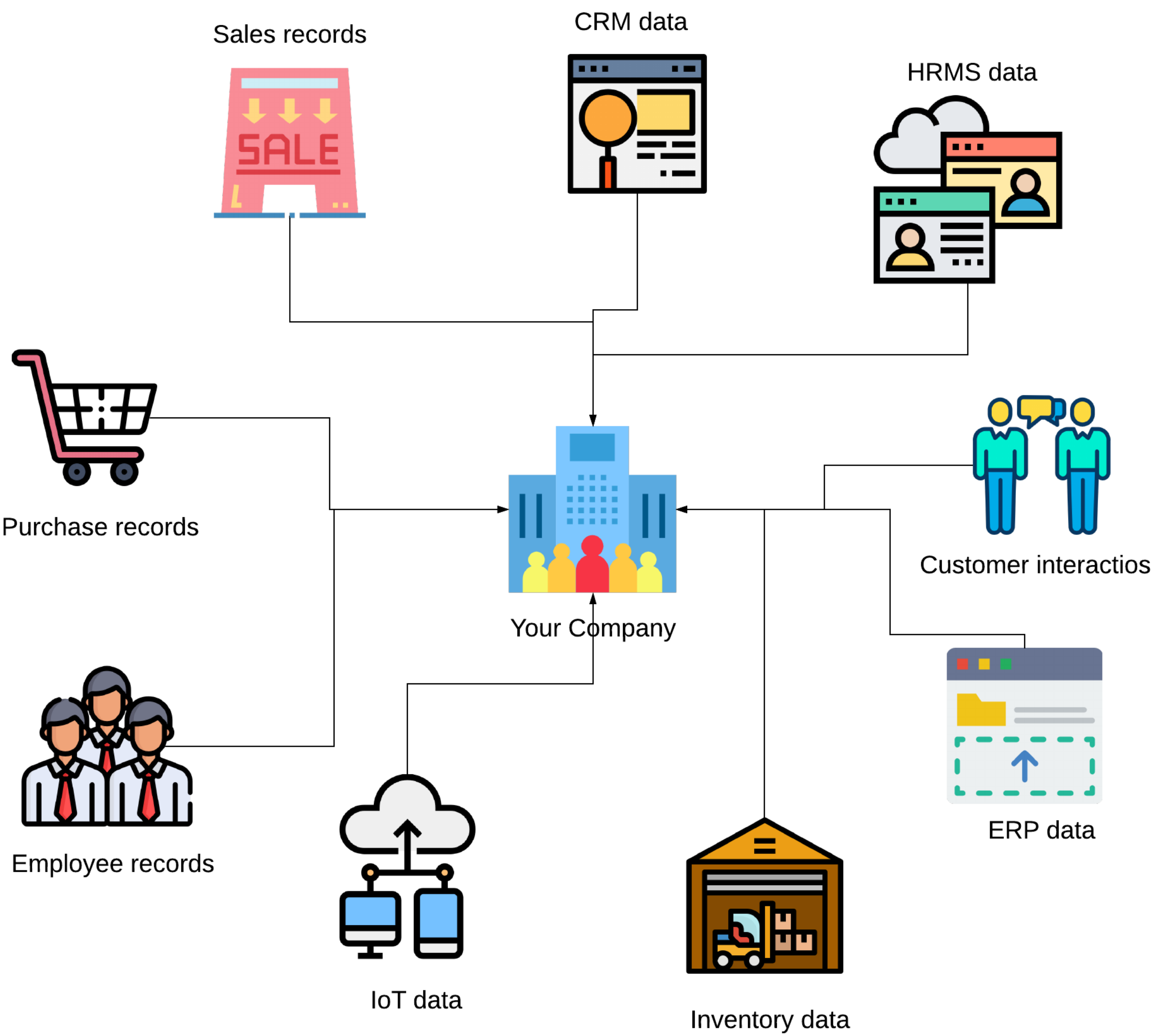
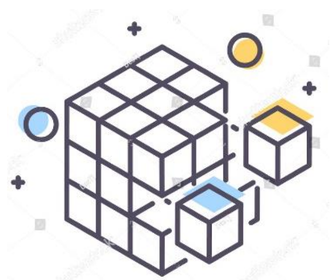


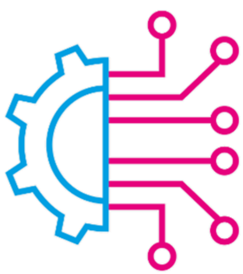
Image credit:
accubits BLOG

Data warehouse

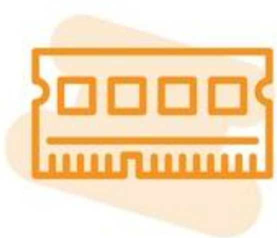
- A data warehouse is a data management system designed to support analytics and decision makings.
- It contains extensive historical data from various sources (e.g., log files and transaction applications).

- The key features of a data warehouse include


Subject-oriented


Integrated


Time-variant


Nonvolatile

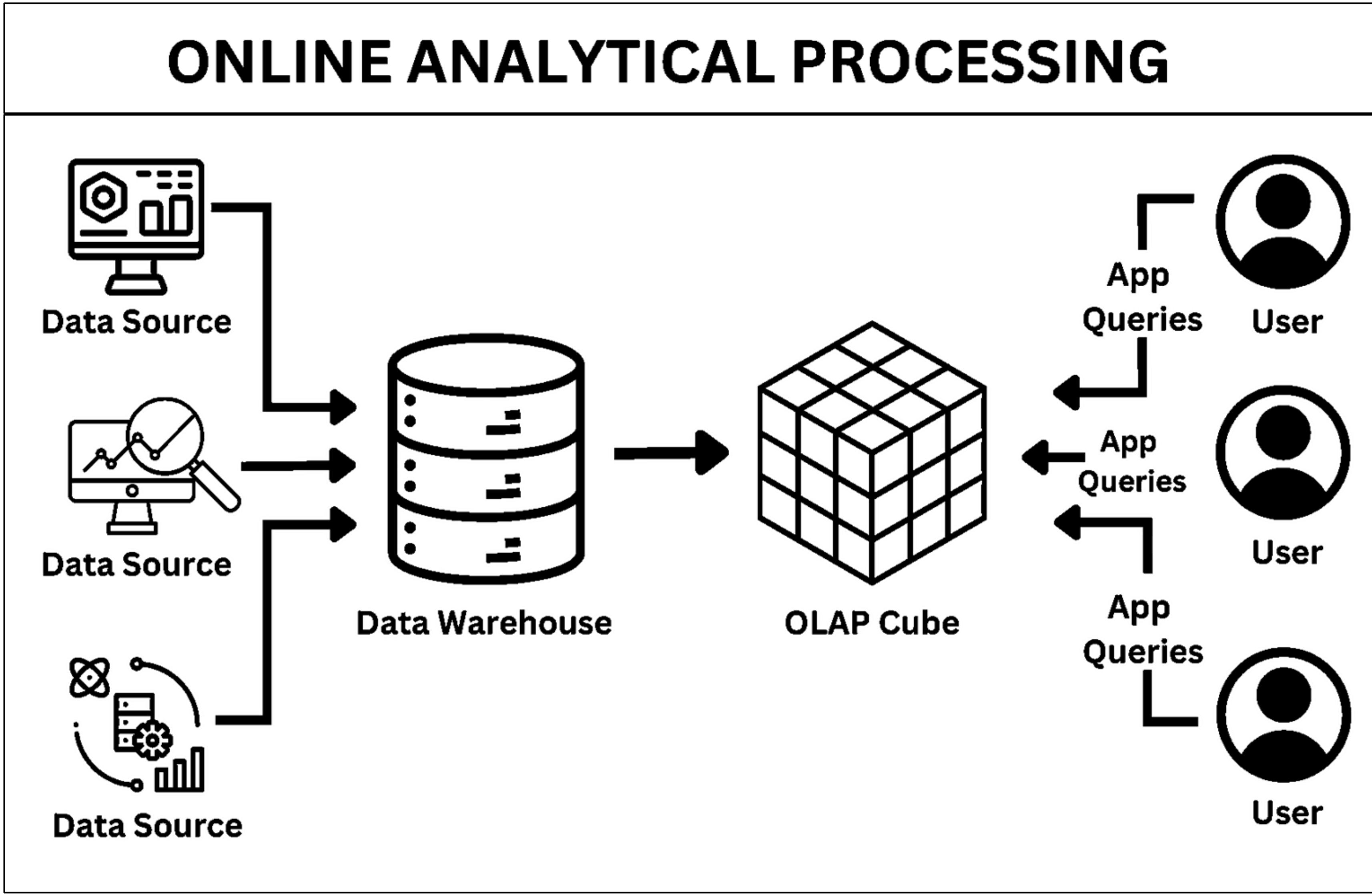
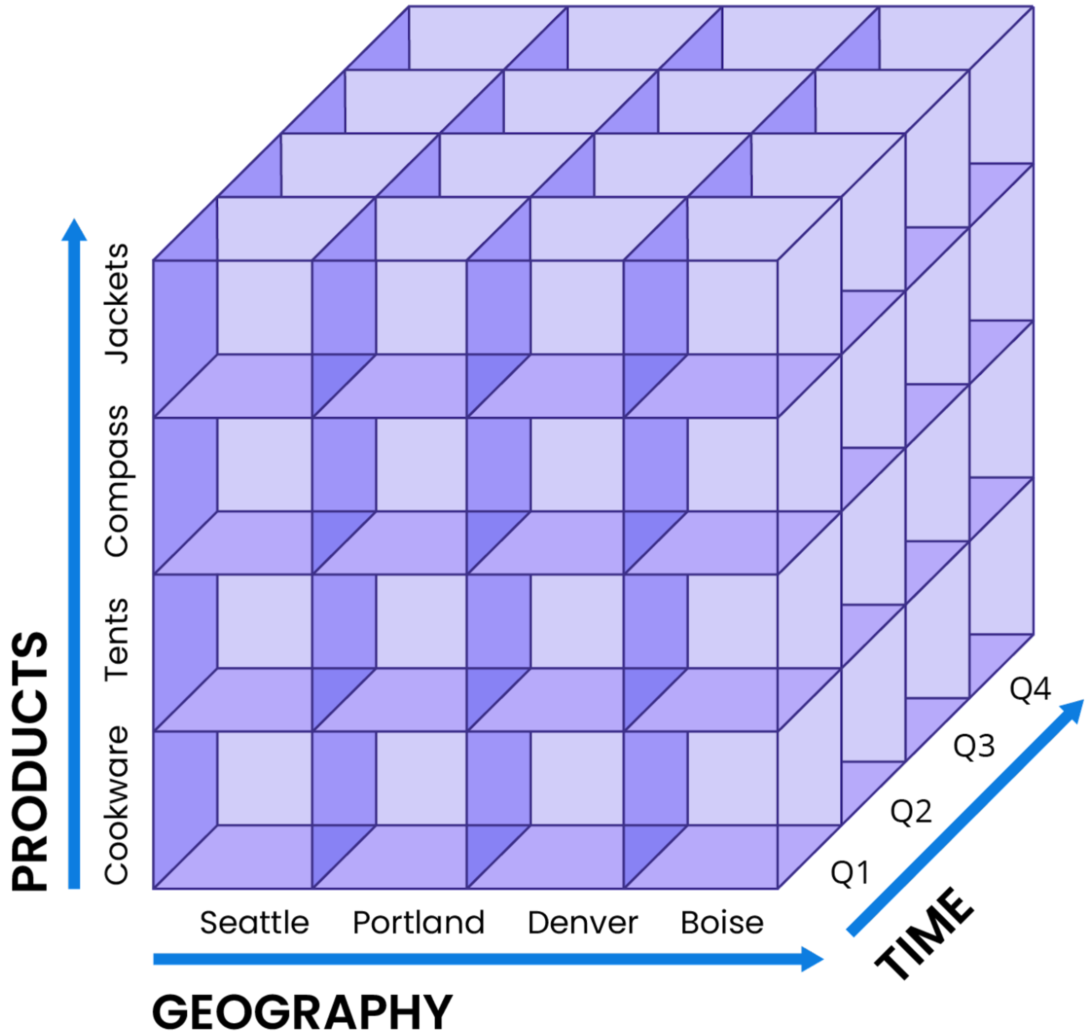
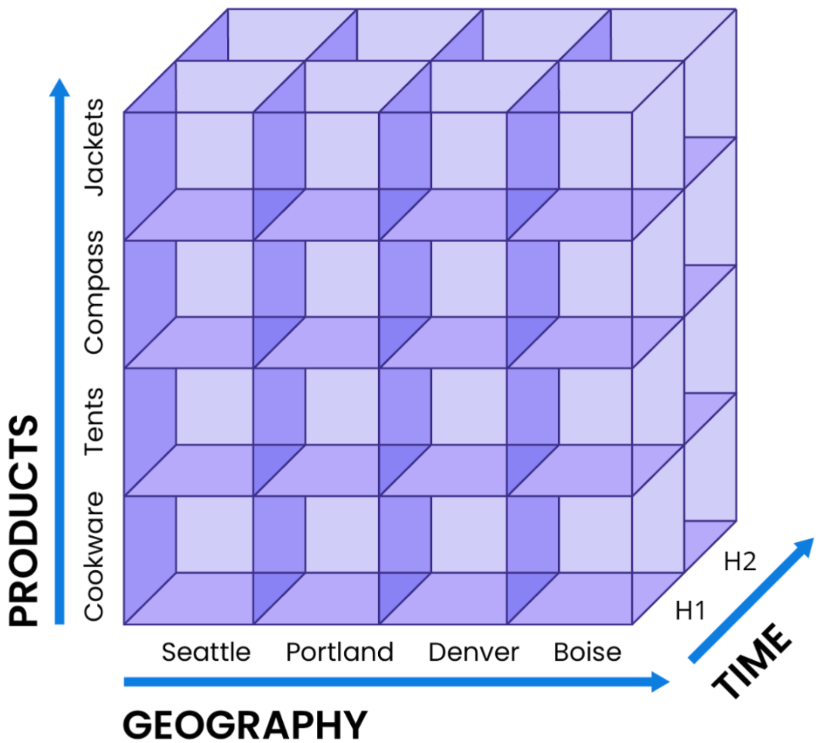


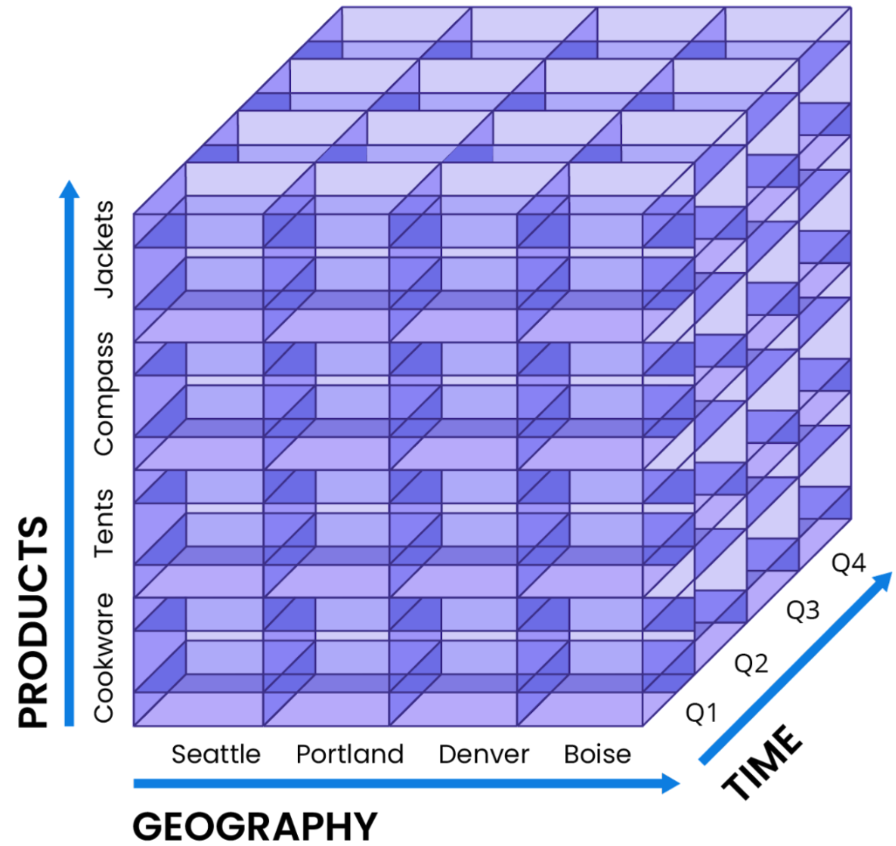
Image credit: [Naveen Chandra blog]



Original data cube

Image credit:
imply.io

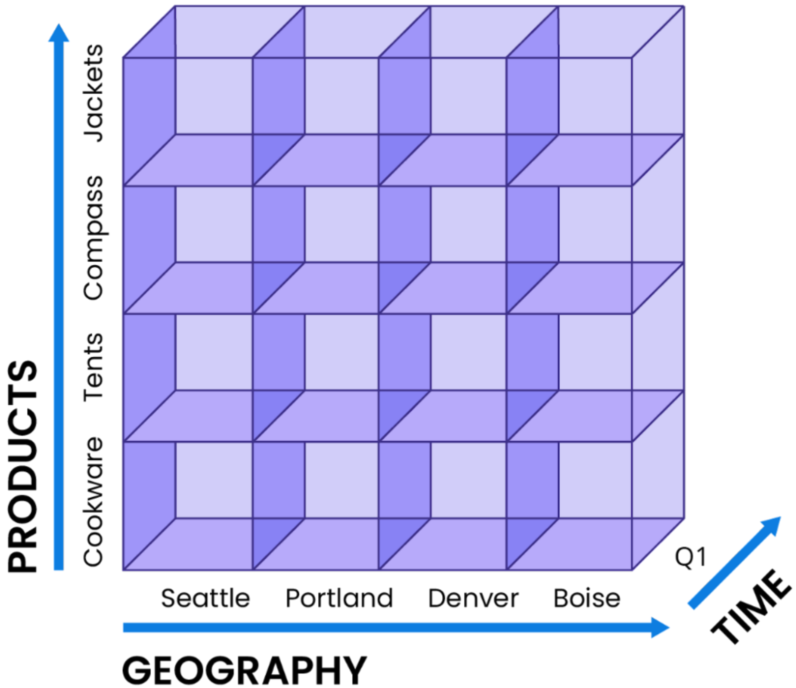




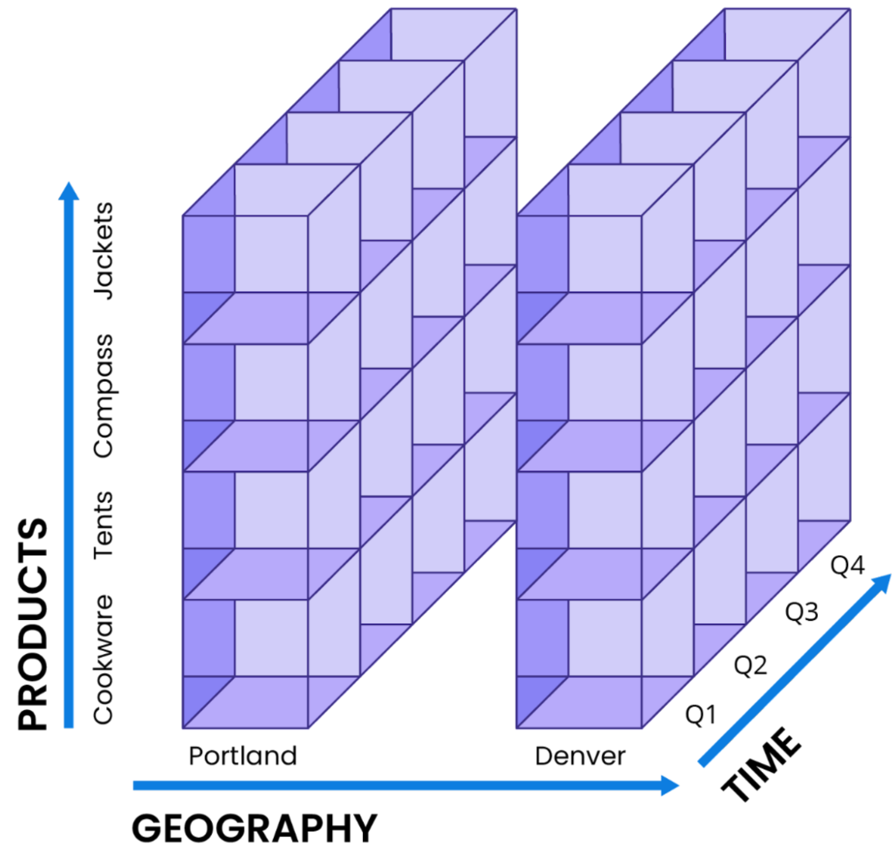
Roll up

Drill down

Image credit:
imply.io

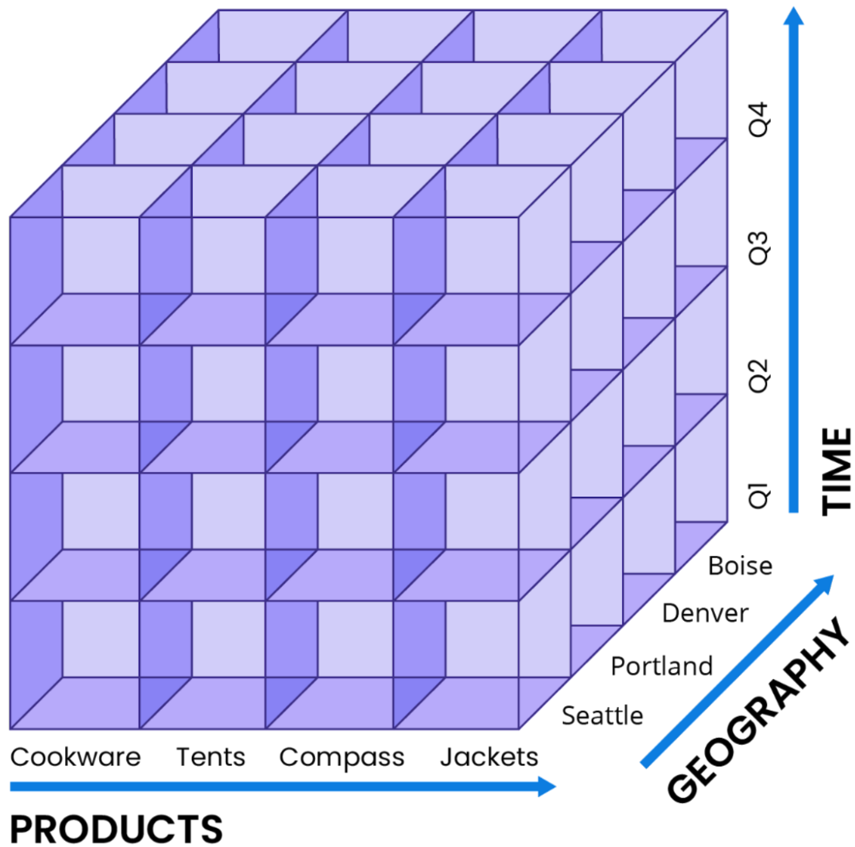


Slice



Dice

Image credit:
imply.io



Pivot

Image credit:
imply.io

OLAP vs. OLTP

Online Analytical Processing (OLAP)	Online Transactional Processing (OLTP)
In-depth data analysis for BI	Day-to-day operations of an organization
OLAP often works with historical data	OLTP usually handles real-time data
Execute complex operations, not as quickly and not on vast data	Read, write, and delete data rapidly on high-volume data
Involve large amounts of parallel users and queries	Not as many users and queries

Data mart

- A data mart functions the same as a data warehouse does but within a much more limited scope.

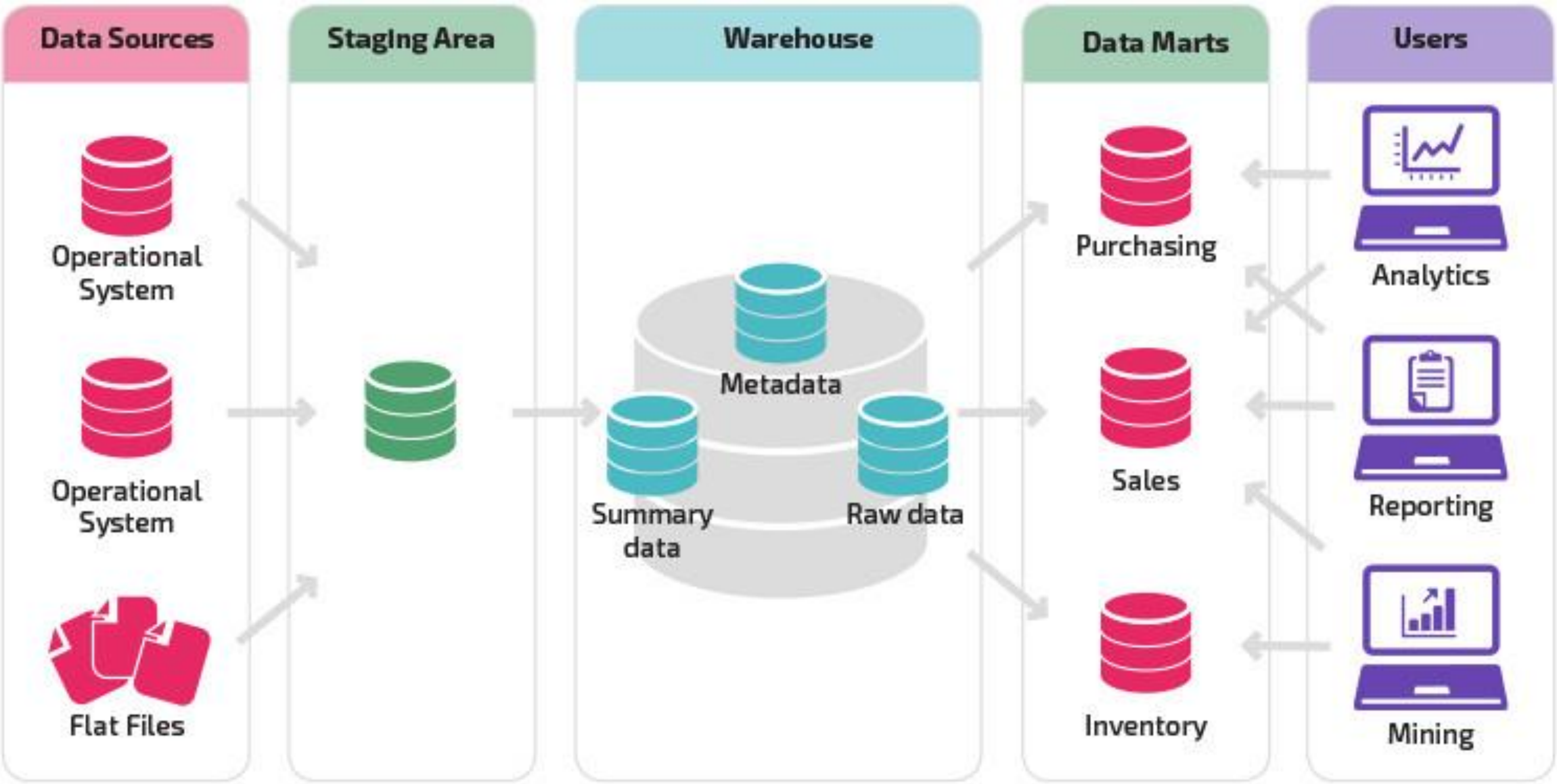


Image credit: Panoply

Data lake

- A data lake is a centralized repository that stores all the data at any scale in native format.

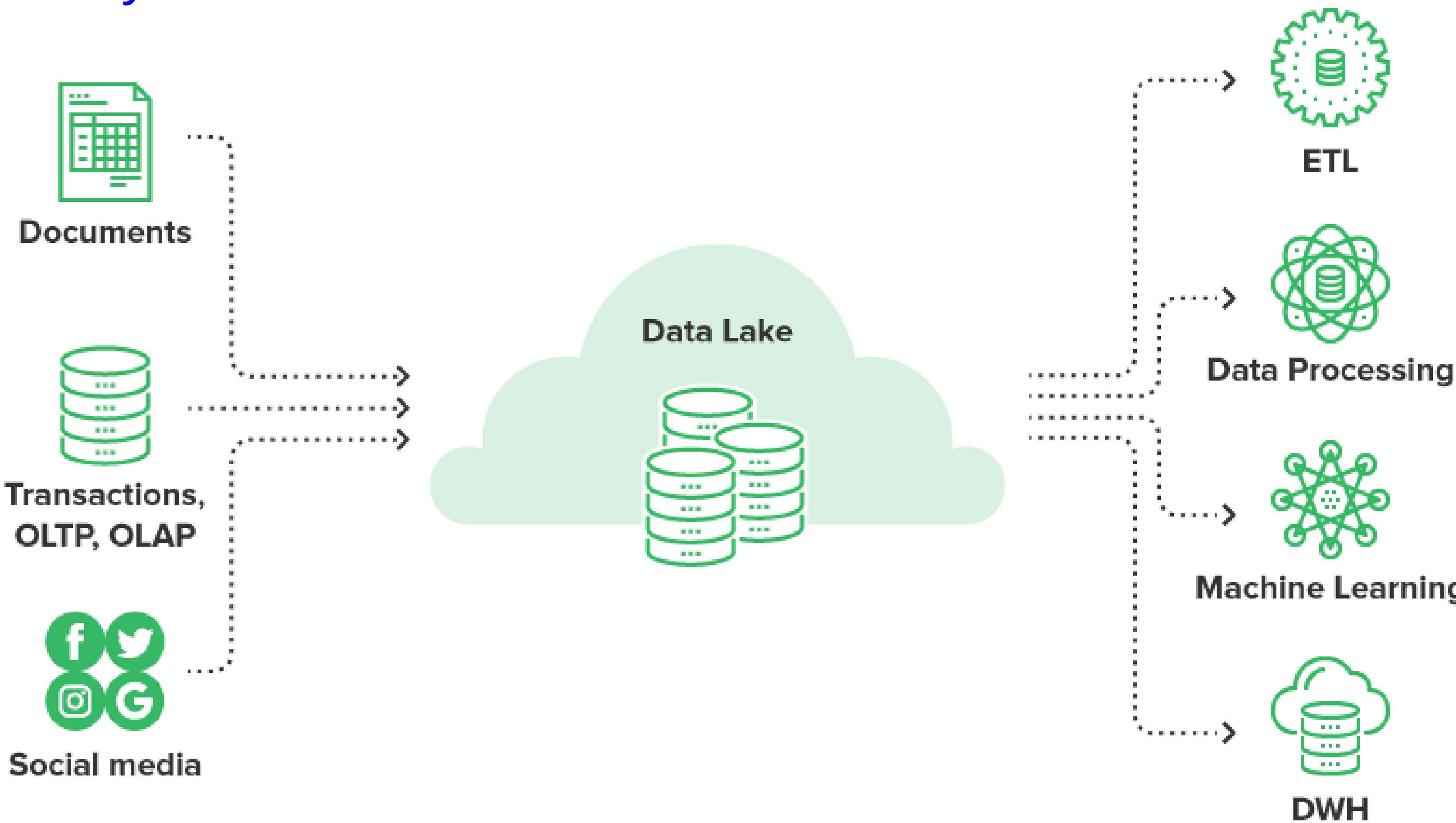


Image credit: CloverDX Blog

Data lake vs. Data warehouse

Characteristics	Data Warehouse	Data Lake
Data	Relational from transactional systems, operational databases, and line of business applications	Non-relational and relational from IoT devices, web sites, mobile apps, social media, and corporate applications
Schema	Designed prior to the DW implementation (schema-on-write)	Written at the time of analysis (schema-on-read)
Price / Performance	Fastest query results using higher cost storage	Query results getting faster using low-cost storage
Data quality	Highly curated data that serves as the central version of the truth	Any data that may or may not be curated (i.e., raw data)
Users	Business analysts	Data scientists, Data developers, and Business analysts (using curated data)
Analytics	Batch reporting, BI and visualizations	Machine Learning, Predictive analytics, data discovery and profiling

Source: aws.amazon.com

Data warehouse tools

teradata.

ORACLE EXADATA

IBM Db2

amazon REDSHIFT

Google BigQuery

snowflake

Data lake tools

AWS Lake Formation

Informatica INTELLIGENT DATA LAKE

infor

Qubole THE OPEN DATA LAKE COMPANY

Azure Data Lake

An example of unified framework

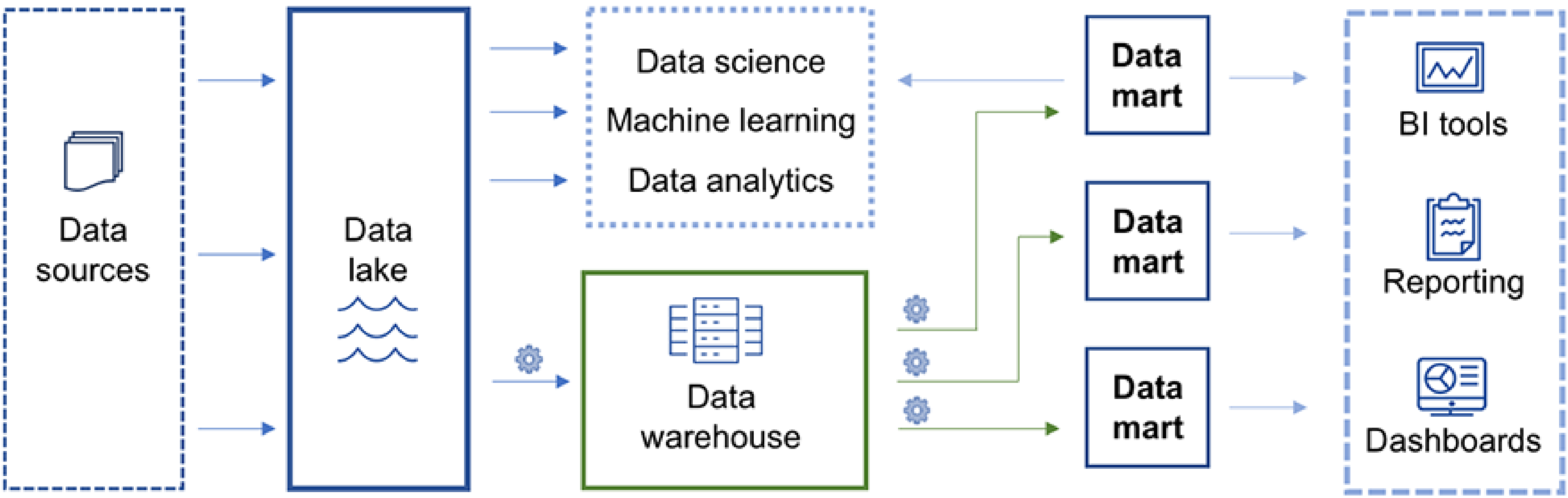
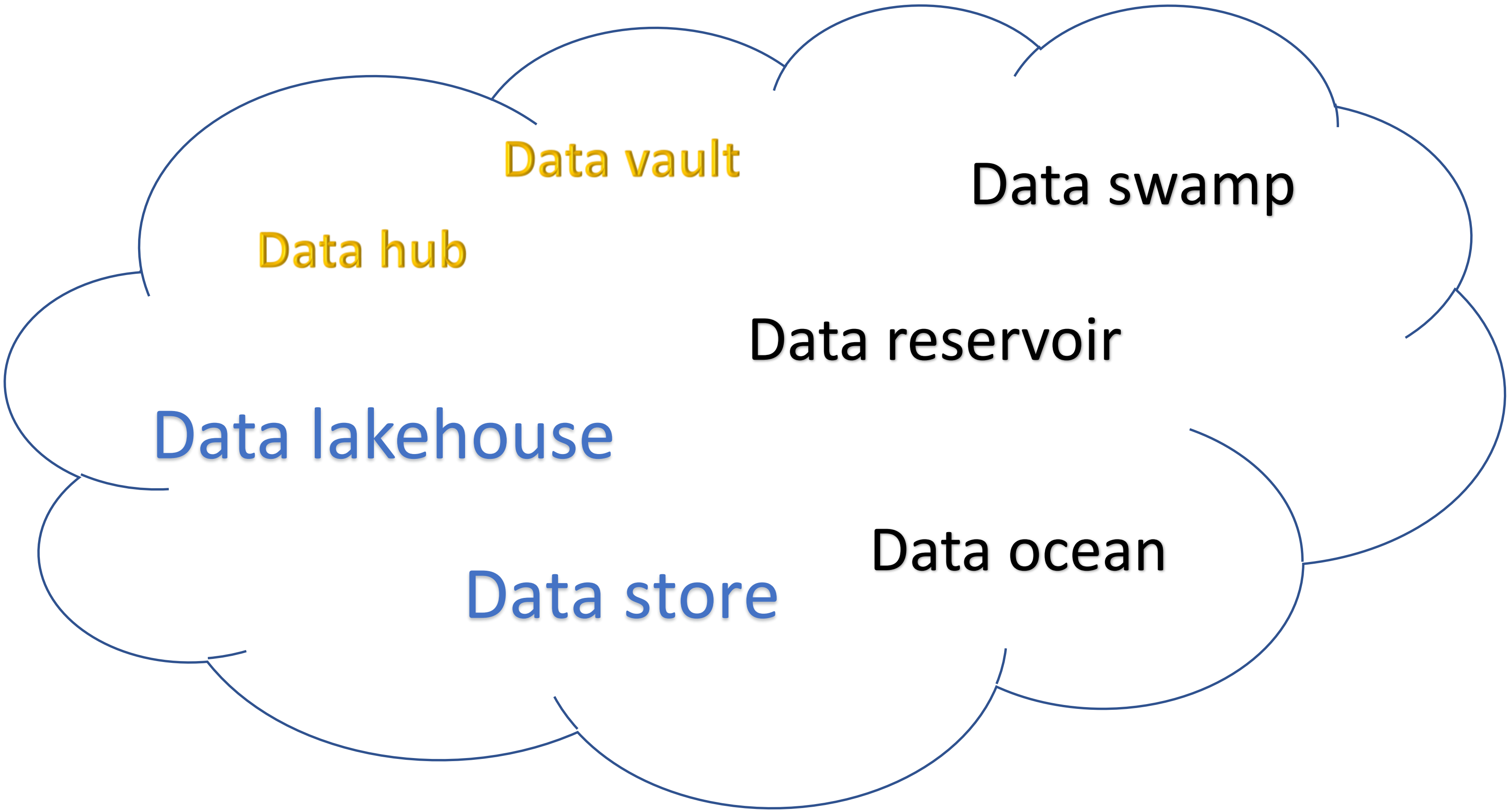


Image credit: Impetus

Have you ever seen these terms? Discover them!



ETL: Extract – Transform – Load

- ETL is the process of moving data from one or multiple data sources to a unified repository.
- Extraction: pull the raw data from different sources
  - Data could come from transactional applications or IoT sensors and in several formats (e.g., relational databases, XML, JSON).
- Transformation: update the data to match organizational needs and data storage solution requirements.
- Loading: deliver and secure the data for sharing, making business-ready data available to internal and external users

ELT: Extract – Load – Transform

- ELT is a variation of ETL in which data is extracted and loaded before it is transformed.

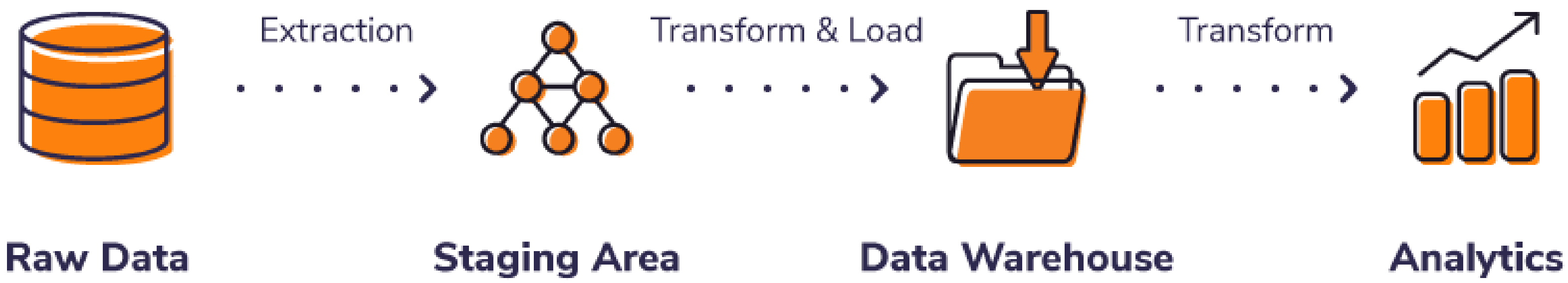
- ELT vs. ETL:

Transform and Load

Data repository

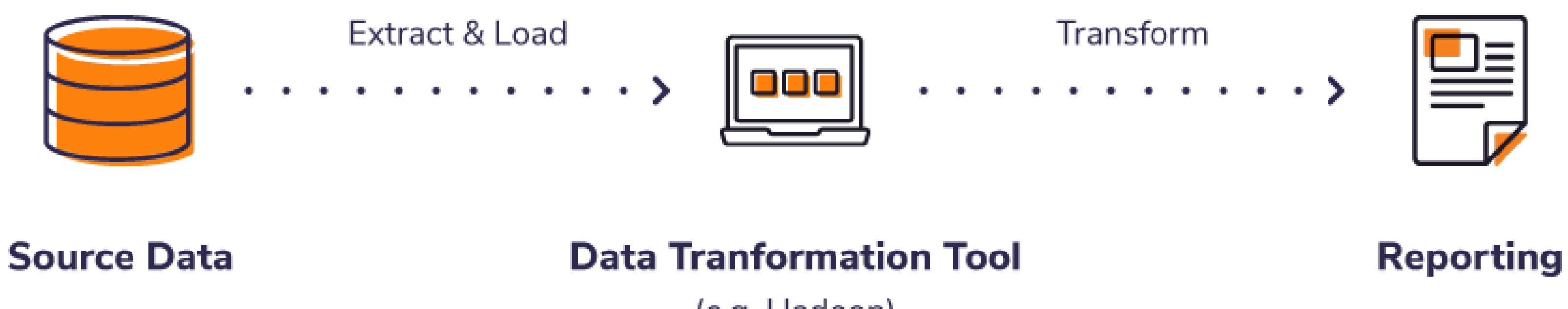
Supporting tools and communities

The ETL Process



Software Advice.

The ELT Process



Data ingestion

- Data ingestion is the process of driving a wide variety of data sets into where it needs to be in required format and quality.
- ETL and ELT are both viable methods, depending on your business needs

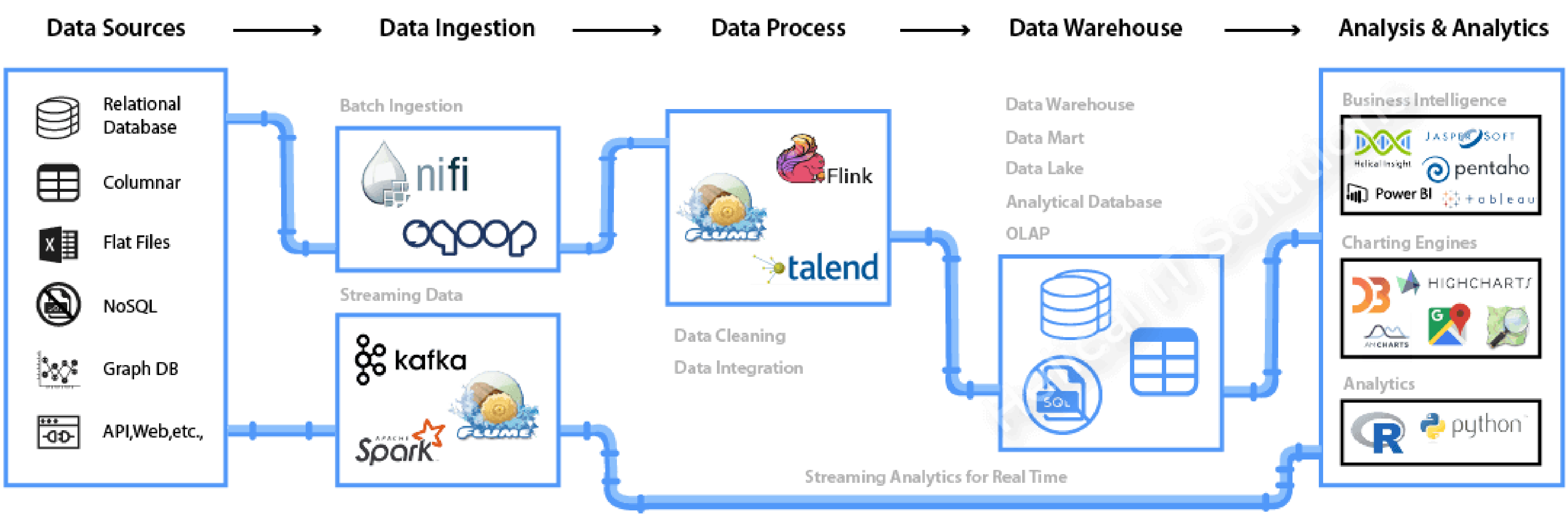
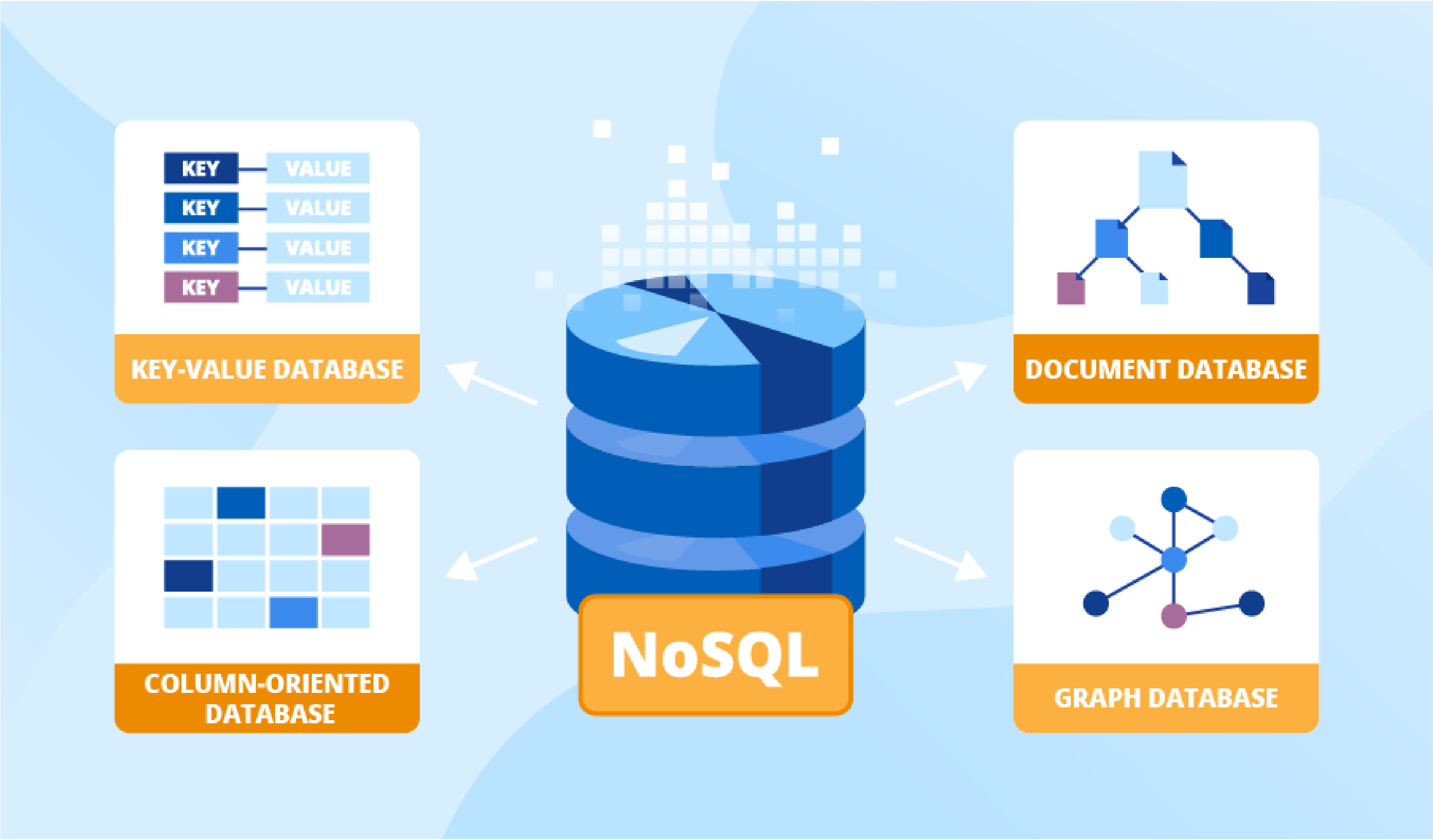


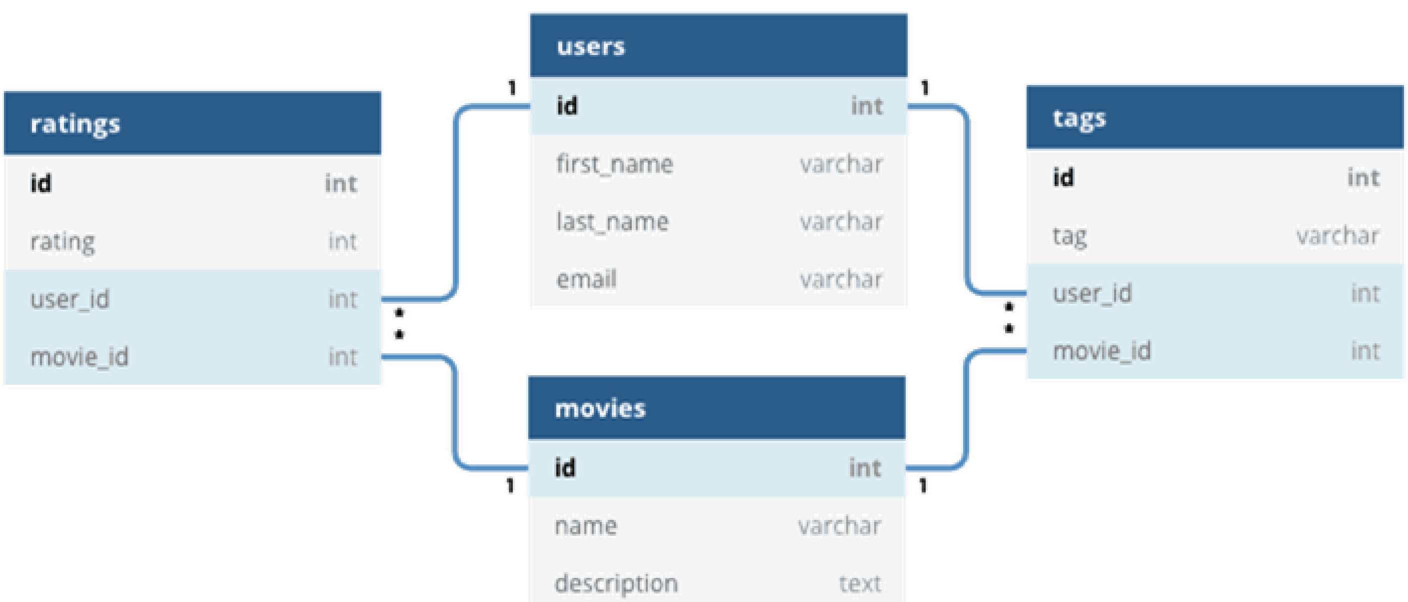
Image credit: Helical IT Solutions Pvt Ltd



NoSQL databases

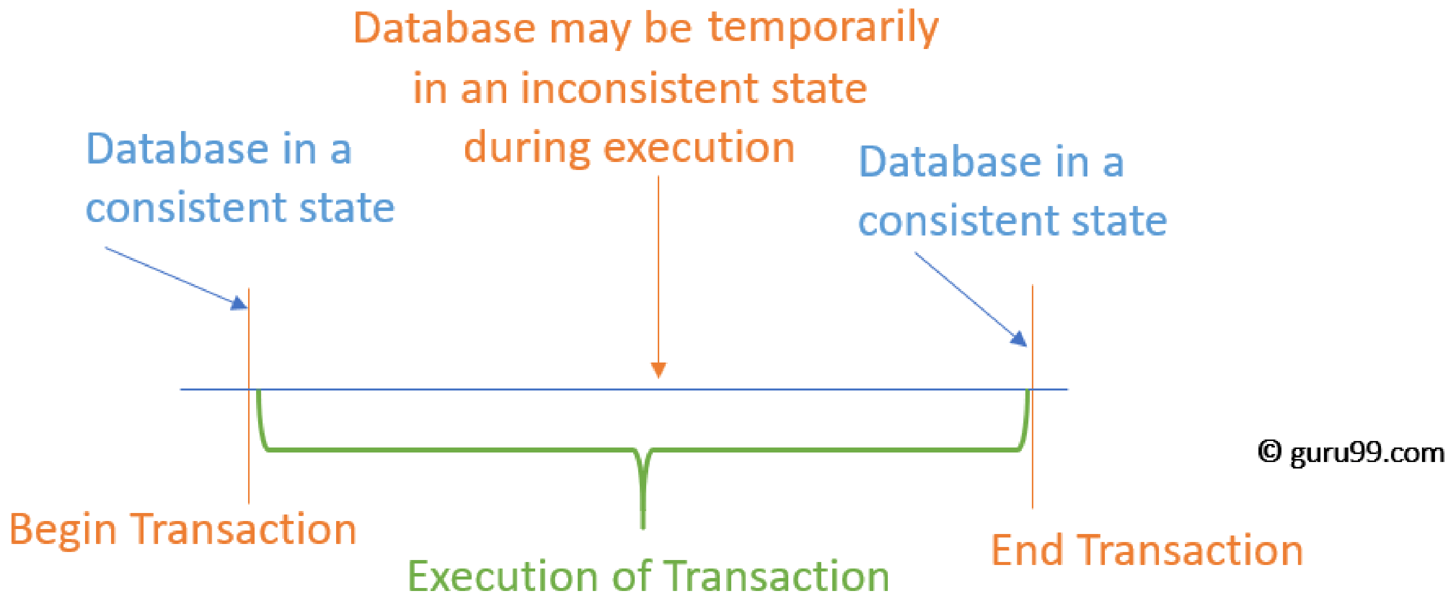
Relational databases

- Relational DB has been a prevalent technology for decades.
  - Mature, proven, and widely implemented
  - Competing database products, tooling, and expertise abound
- Related tables are managed in a strict schema using SQL.
- ACID guarantees are supported.



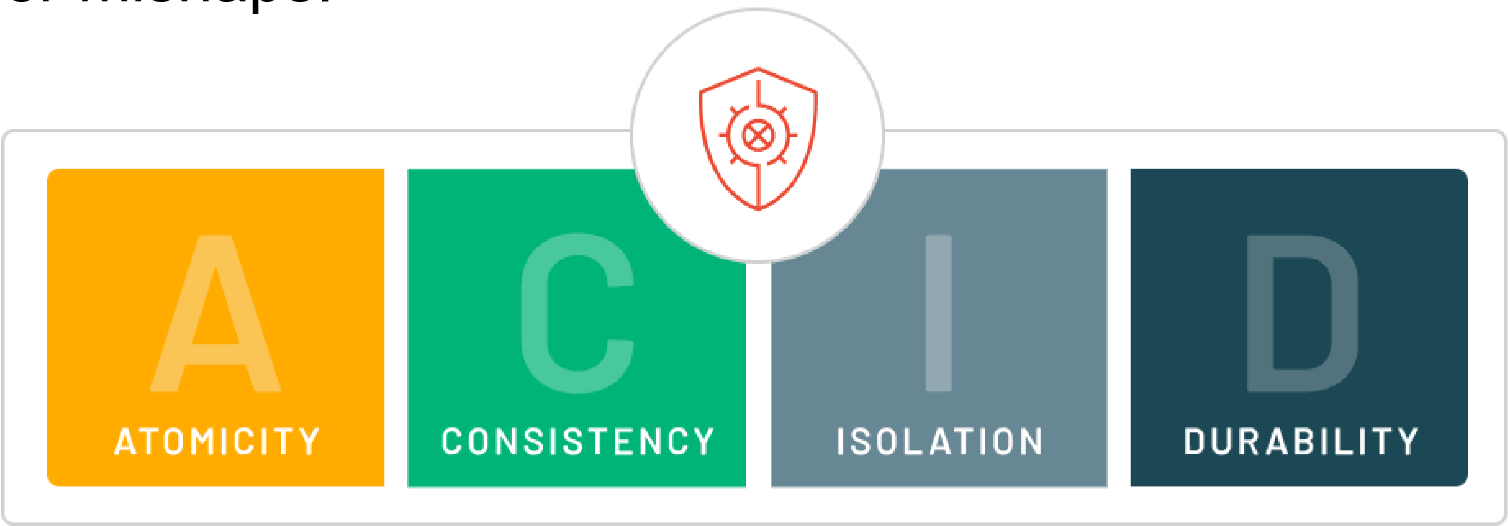
Transactions in RDBMS

- A transaction is any operation treated as a single unit of work, which either completes fully or does not complete at all.
  - E.g., withdraw money from bank account: either the money has left your bank account, or it has not – there cannot be an in-between state
- It leaves the storage system in a consistent state.



ACID properties

- This is a set of properties for database transactions.
- They ensure data validity despite errors, power failures, and other mishaps.



- Data storage systems that apply these operations are called transactional systems.

Atomicity

- A transaction is a single unit of operation. You either execute it entirely or do not execute it at all. There cannot be partial execution.
- This property prevents data loss and corruption from occurring.

Consistency

- Transactions only make changes to tables in predefined and predictable ways.
- Data errors or corruption do not create unintended effect for the integrity of tables.

Isolation

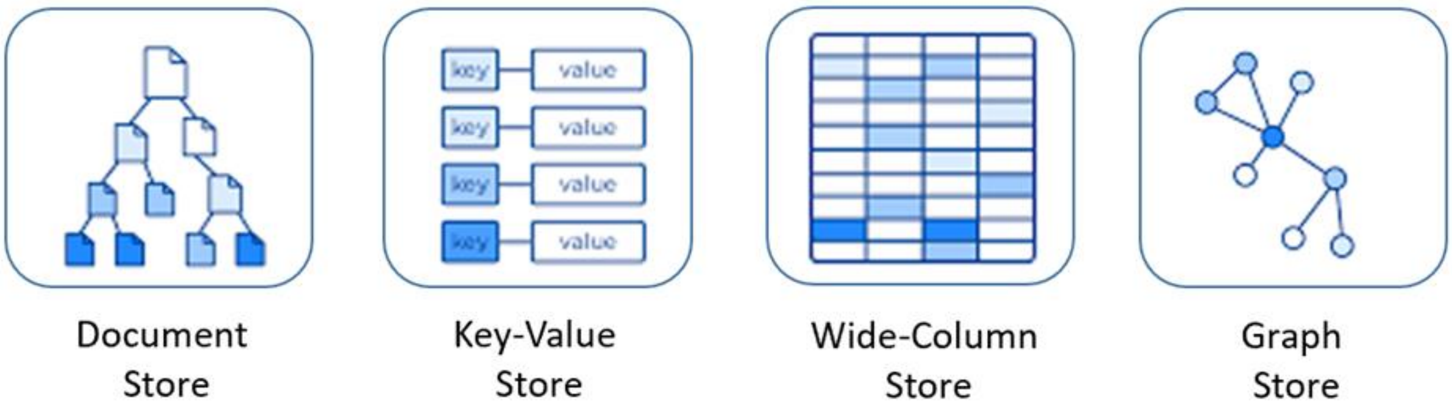
- The concurrent transactions of multiple users do not interfere with another.
- During execution, intermediate transaction results from simultaneously executed transactions should not be made available to each other.

Durability

- Changes to data made by successfully executed transactions will be saved, even in the event of system failure.

What are NoSQL databases?

- NoSQL databases are non-tabular and they store data in a format other than relational tables.
- They come in a variety of types based on their data models.



- Provide flexible schemas
- Scale easily with large amounts of data and high user loads

NoSQL databases

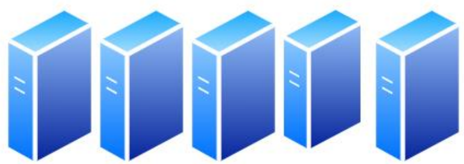
- NoSQL databases excel in their scalability, resilience, ease-of-use, and availability characteristics.
  - They are appropriate for high-volume services that need sub-second response time.
- They typically do not support ACID guarantees beyond the scope of a single database partition.
- Applications

Gaming | Social network | IoT | Web | Mobile | Enterprise

NoSQL databases: Key features

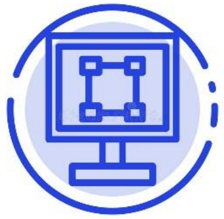
- Each NoSQL database has its own unique features.
- At a high level, many NoSQL DBs have the following features

Flexible schema



Horizontal scaling

Fast queries due to the data model

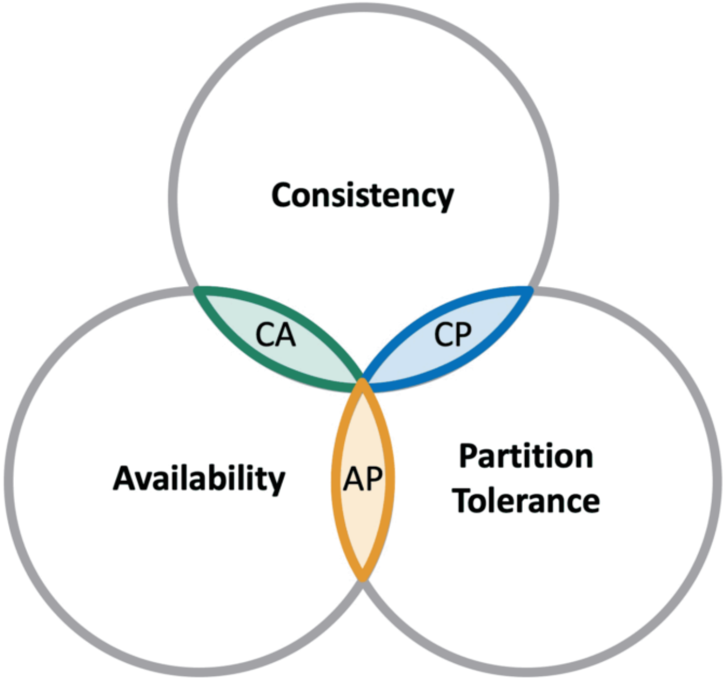


Easy development

Consider a NoSQL datastore when:	Consider a relational database when:
The workload volume is high. It requires predictable latency at large scale (e.g., latency measured in milliseconds while performing millions of transactions per second)	The workload volume generally fits within thousands of transactions per second
Data is dynamic and frequently changes	Data is highly structured and requires referential integrity
Relationships can be de-normalized data models	Relationships are expressed through table joins on normalized data models
Data retrieval is simple and expressed without table joins	You work with complex queries and reports
Data is typically replicated across geographies and requires finer control over consistency, availability, and performance	Data is typically centralized, or can be replicated regions asynchronously
Your application will be deployed to commodity hardware, e.g., with public clouds	Your application will be deployed to large, high-end hardware

The CAP theorem

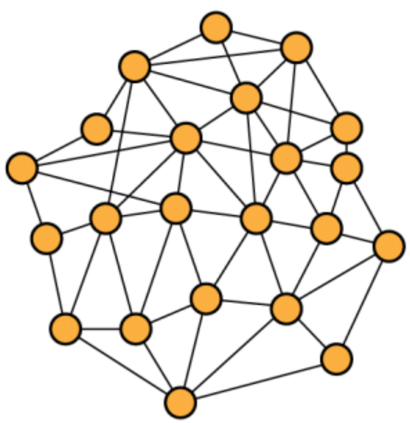
- The CAP theorem describes a set of principles applied to distributed systems that store state.



- It explains the tradeoffs native to managing consistency and availability during a network partition.
  - However, tradeoffs with respect to consistency and performance also exist with the absence of a network partition.

C – Consistency

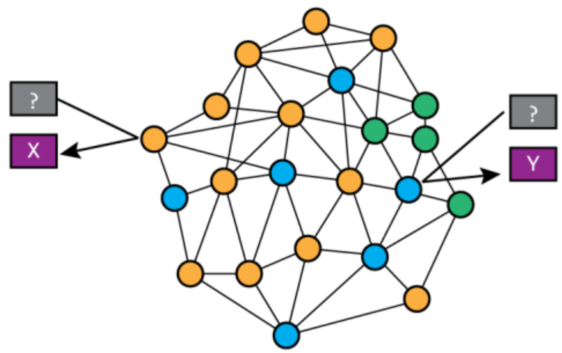
At any given time, all nodes in the network have exactly the same (most recent) value.



= Value: X @ 2018-05-03T08:52:40

A – Availability

Every request to the network receives a **response**, without any guarantee that returned data is the most recent.



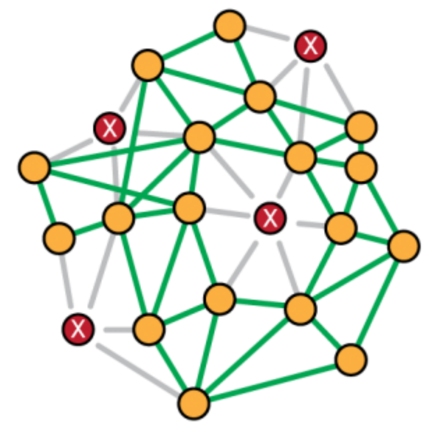
= Value: X @ 2018-05-03T08:52:40

= Value: Z @ 2018-05-03T08:32:58

= Value: Y @ 2018-05-03T07:12:12

P – Partition Tolerance

The network continues to operate, even if an arbitrary number of nodes are **failing**.

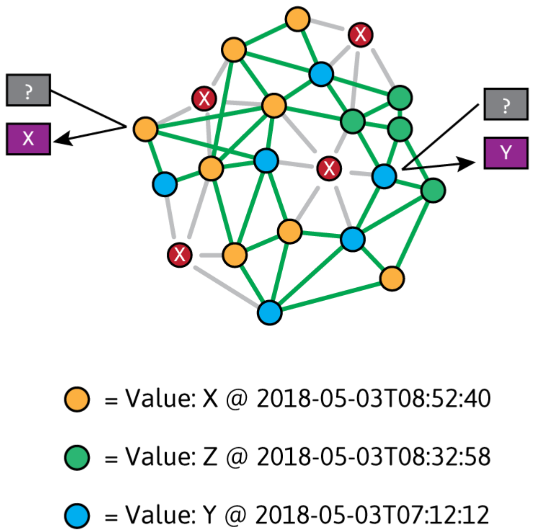


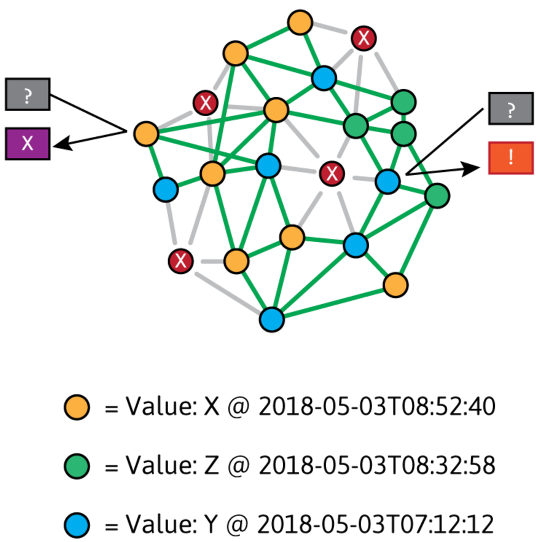
The CAP theorem

- Any database can only guarantee two of the three factors.
  - E.g., RDBMS (CA), MongoDB (CP), and Cassandra (AP)
- Relational databases are CA but not P.
  - They are typically provisioned to a single server and scale vertically by adding more resources to the machine.
- NoSQL databases are typically CP or AP.
  - Partition tolerance can never be ruled out $\rightarrow$ there will always be failing nodes in the network.
  - One must choose between C and A when in the presence of P.

Distributed systems with CAP

- Availability over Consistency (A + P)
  - Every request receives a response, even if the network cannot guarantee it is up to date due to network partitioning (failing nodes).
  - The system will be highly available, yet not guarantee that the data returned is most recent.





- Consistency over Availability (C + P)
  - The system will return an error or a time-out if information cannot be guaranteed to be up to date due to network partitioning.
  - The system will be highly accurate, yet most likely be unavailable for 99.99% of the time.

NoSQL databases: Misconceptions

- NoSQL databases do not store relationship data well.
- They just store relationship data differently than RDBMS do.

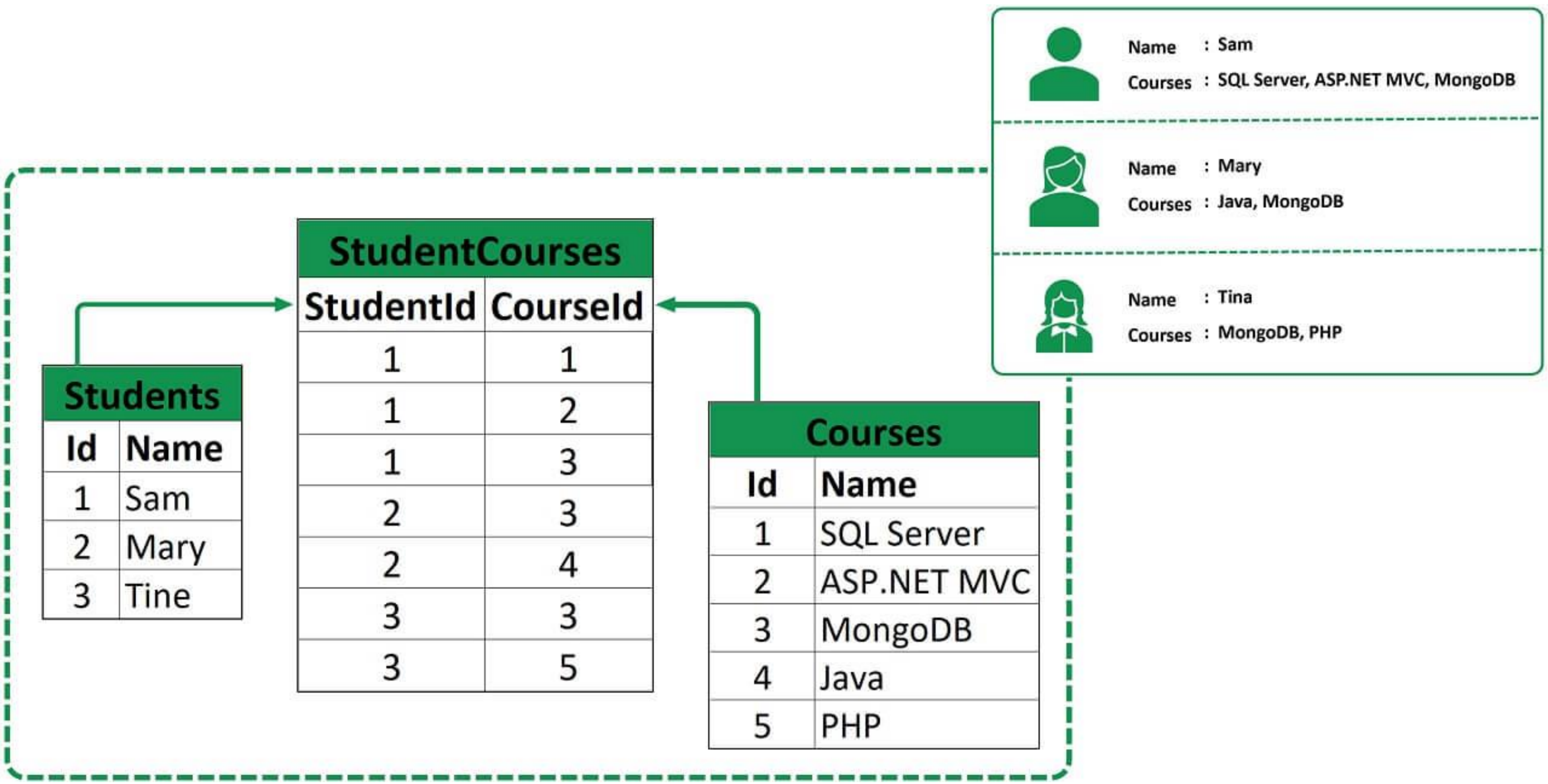


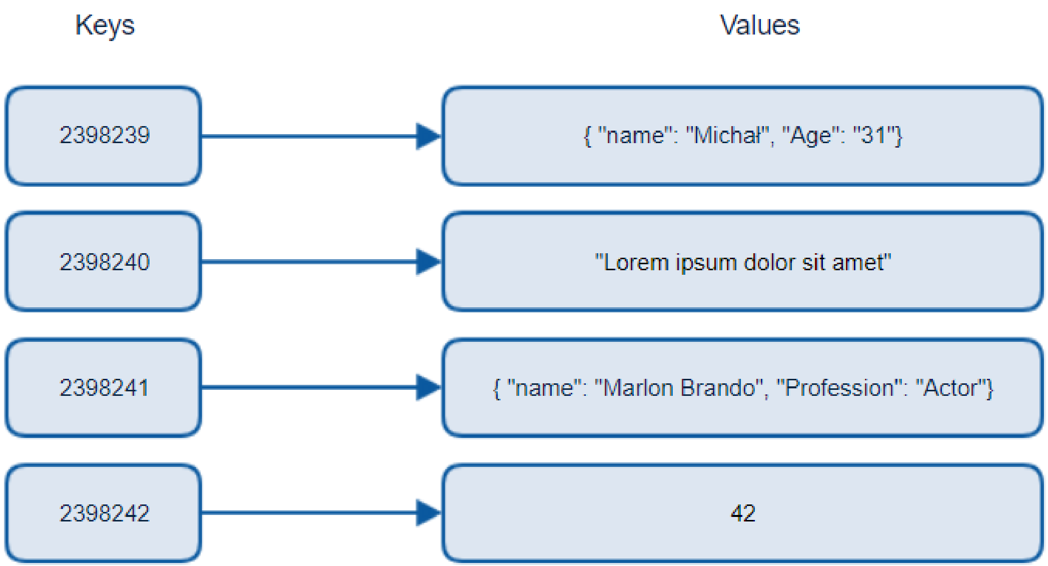
Image credit: [Pragim technology]()

NoSQL databases: Misconceptions

- NoSQL databases do not support ACID transactions.
- Many NoSQL databases do, maybe not fully support.
- The way data is modeled can eliminate the need for multi-record transactions in many use cases.

Data models: Key-value store

- The simplest type of NoSQL databases
- The data is represented as a collection of key-value pairs.



Keys and values in Azure Cosmos DB

riak

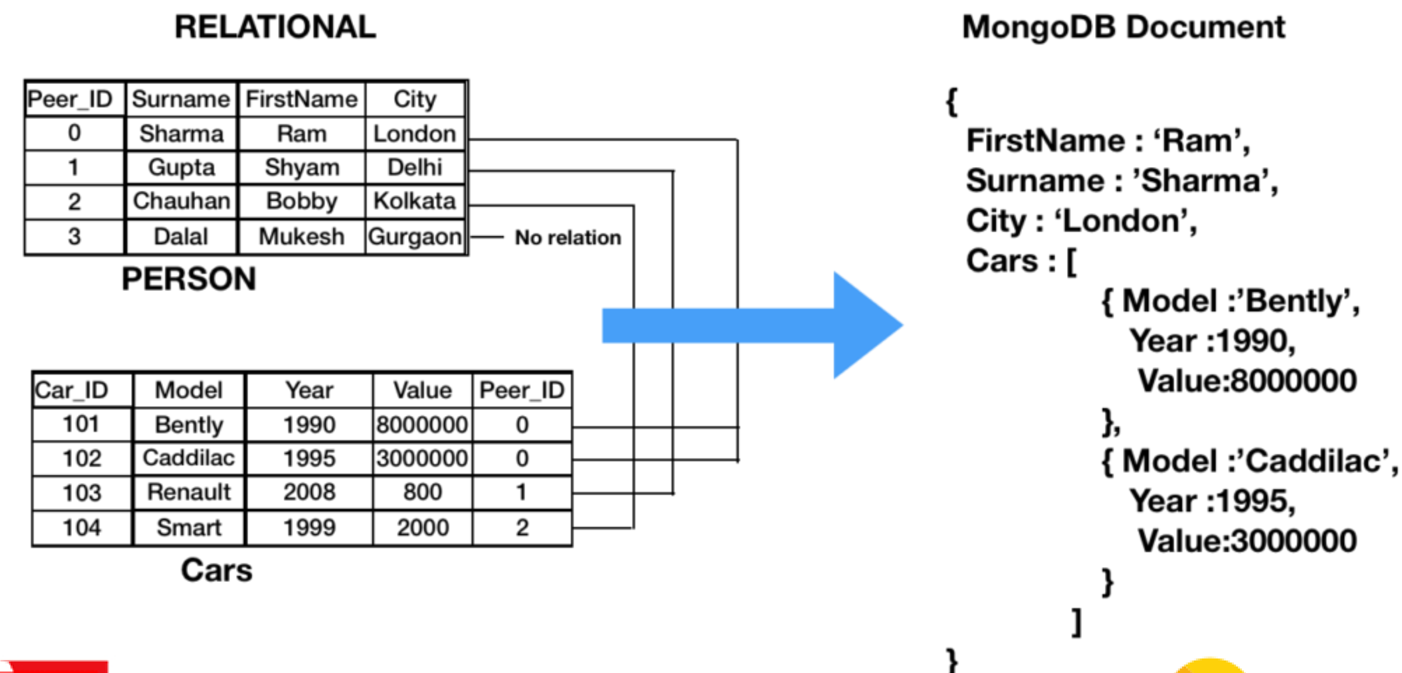
redis

amazon DynamoDB

AEROSPIKE

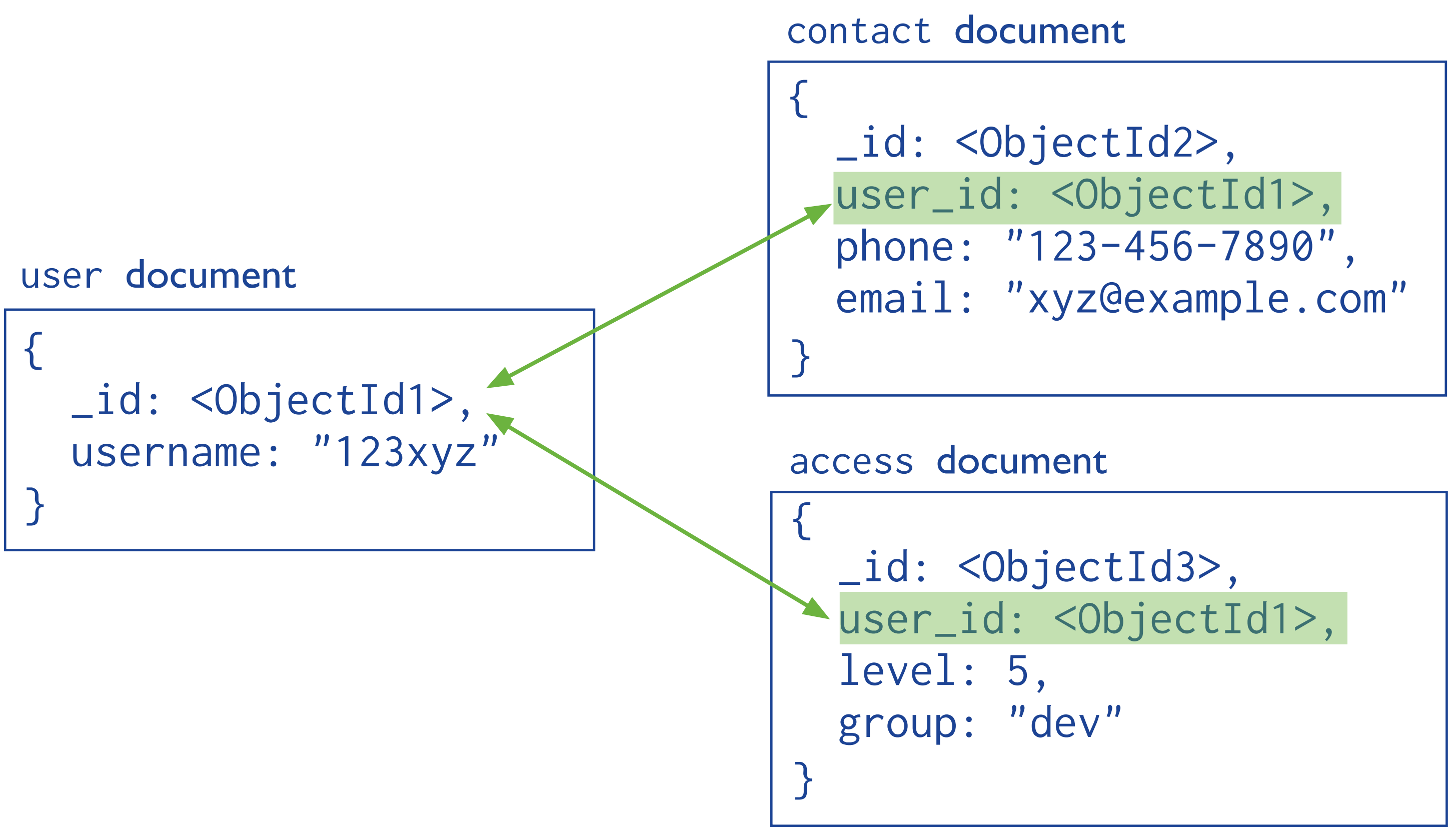
Data models: Document store

- These databases store the data and metadata hierarchically in JSON-like documents.



Normalized data models

- Relationships are depicted by references between documents.



Denormalized data models

- Related data is embedded in a single structure or document.
- This involves simple operations to manipulate related data.

```
{
  _id: <ObjectId1>,
  username: "123xyz",
  contact: {
             phone: "123-456-7890",
             email: "xyz@example.com"
           },
  access: {
            level: 5,
            group: "dev"
          }
}
```

Embedded sub-document

Embedded sub-document

Data models: Wide-column store

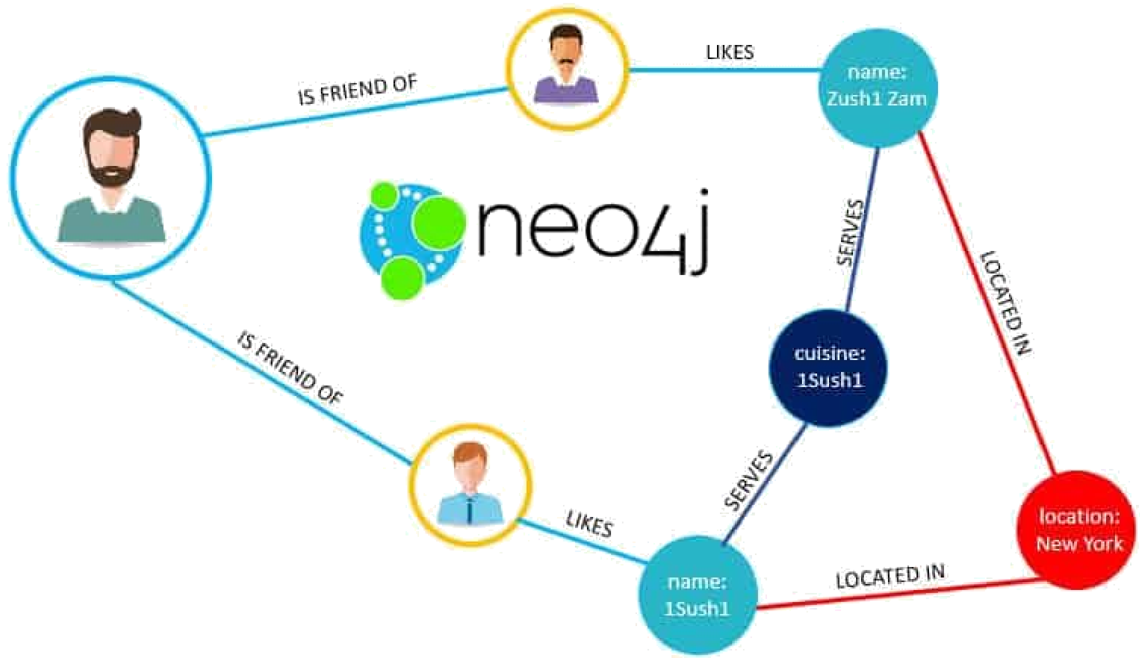
- The data is stored as a set of nested-key/value pairs within a single column.

Row A	Column 1	Column 2	Column 3	...
	Value	Value	Value	

Row B	Column 2	Column 3	Column 4	...
	Value	Value	Value	

Data models: Graph store

- We use graph structures (e.g., node, edge, and properties) to describe the data.



neo4j

JanusGraph

TigerGraph

ArangoDB

AllegroGraph

Multi-model databases

- Multi-model databases are designed to support multiple data models against a single, integrated backend.

	Key-value	Document	Graph	Streaming
Franz Inc. AllegroGraph	☐	☑	☑	☐
ArangoDB	☑	☑	☑	☐
redis	☑	☑	☑	☑

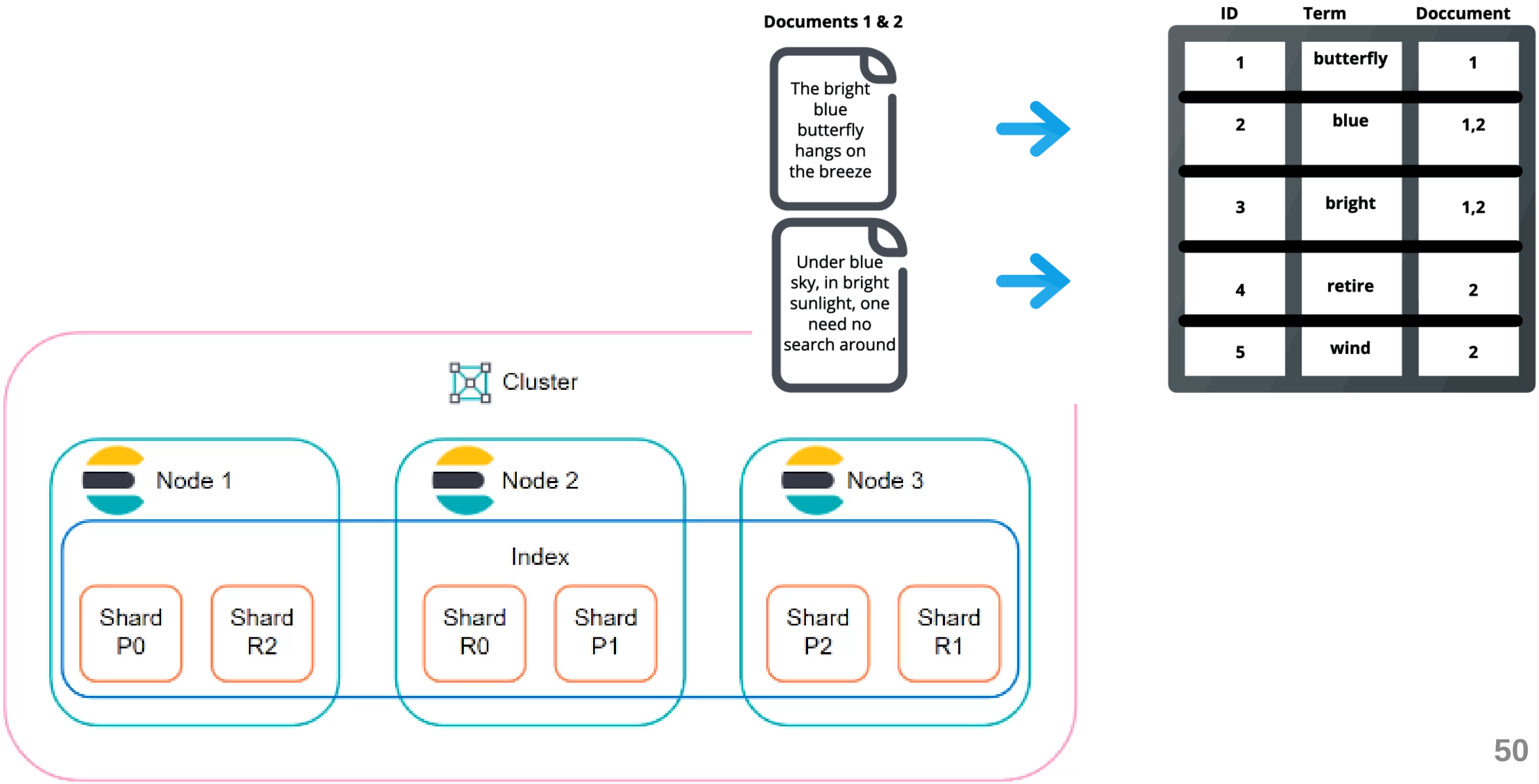
In-memory databases

- In-memory databases primarily store data on main memory, which contrasts to those using physical disks.
- Minimal response time is enabled by eliminating the need to access disks.

A distributed, multitenant-capable full-text search engine with an HTTP web interface and schema-free JSON documents

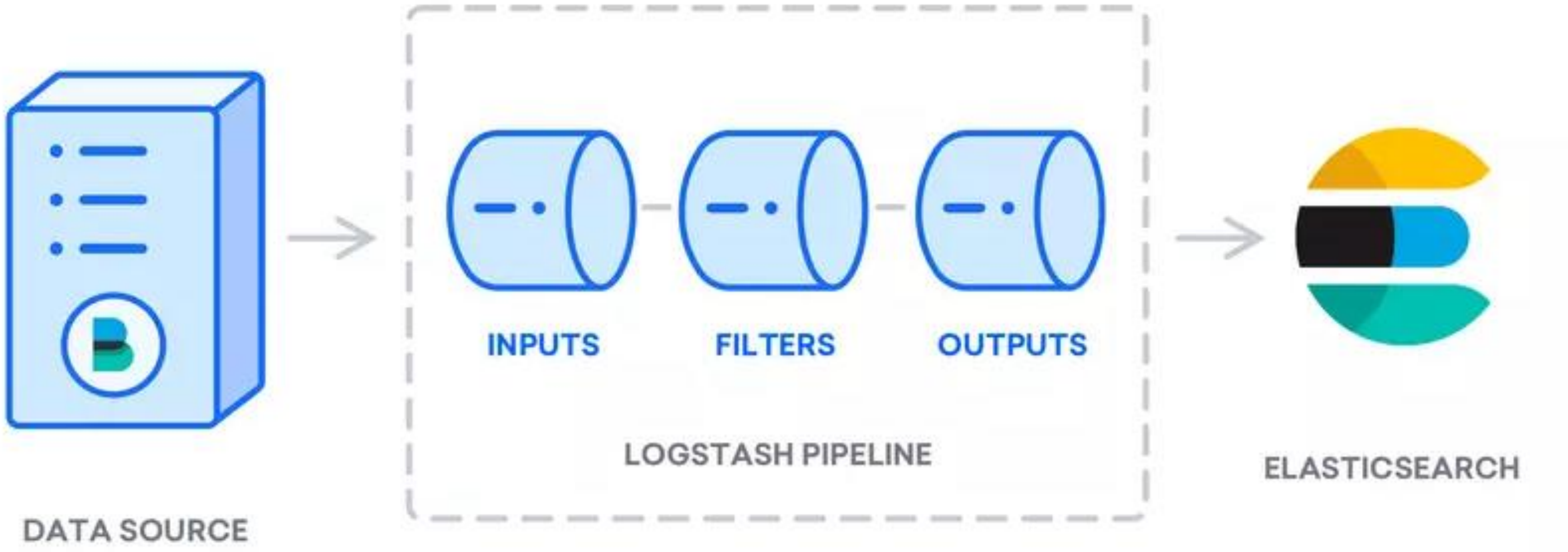
Elasticsearch cluster

- The power of an Elasticsearch cluster lies in the distribution of tasks, searching, and indexing, across all the nodes.



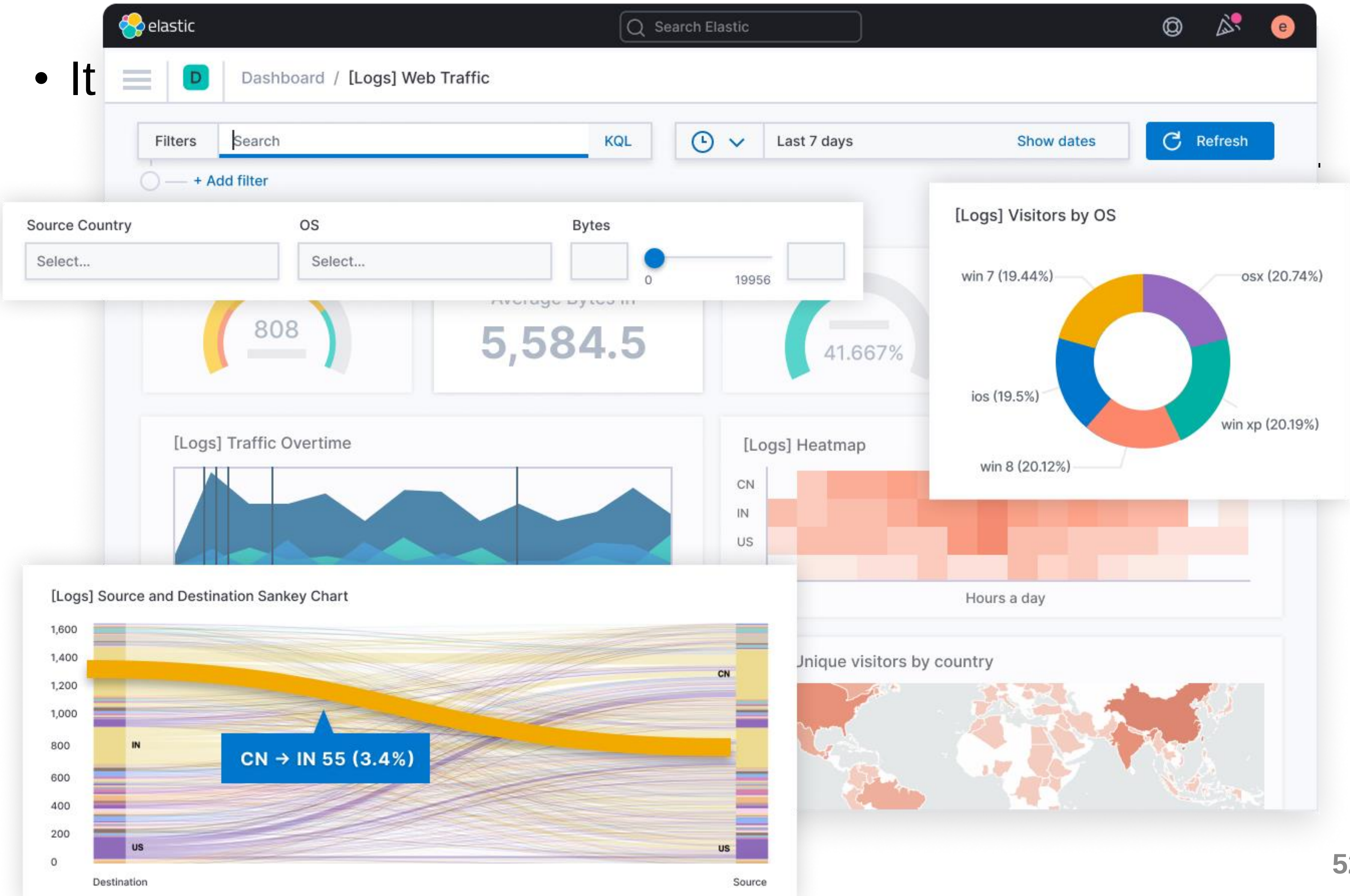
Elasticsearch: Logstash

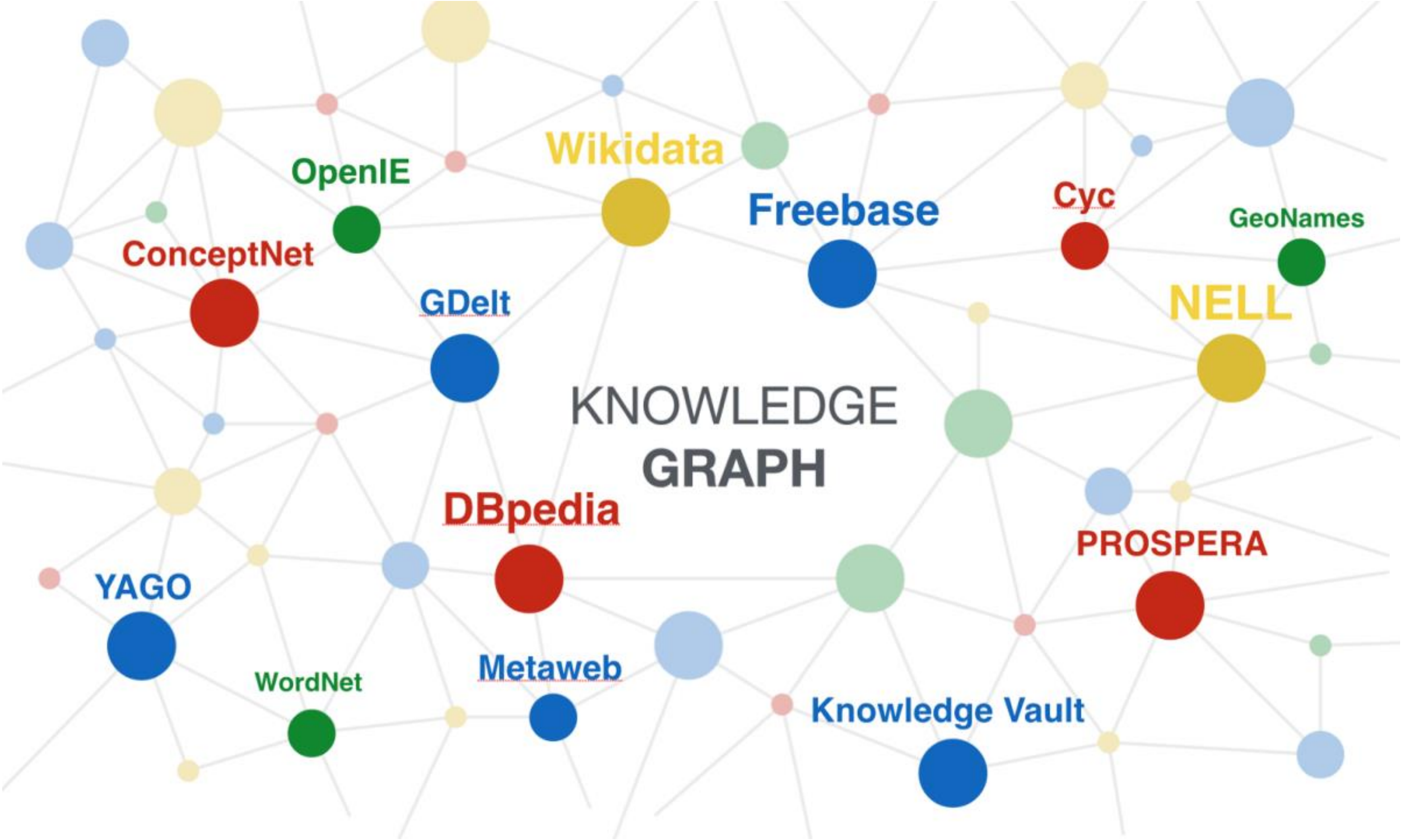
- Logstash ingests data from various sources, transforms it, and sends it to Elasticsearch.



Elasticsearch: Kibana

- It





Graph data: Sources and storage

Graph data sources

[Stanford Network Analysis Platform (SNAP)]

- A wide variety of graphs representing social networks, citation networks, online communities, and more

[Wikidata]

- A collaboratively edited multilingual knowledge graph of 96,886,330 data items

[Yet Another Great Ontology (YAGO)]

- Augment WordNet with common knowledge facts from Wikipedia
- Originally consisted of more than 1 million entities and 5 million facts

Graph data sources

Open Graph Benchmark

- A collection of realistic, large-scale, and diverse benchmark datasets for machine learning on graphs

TUDatasets

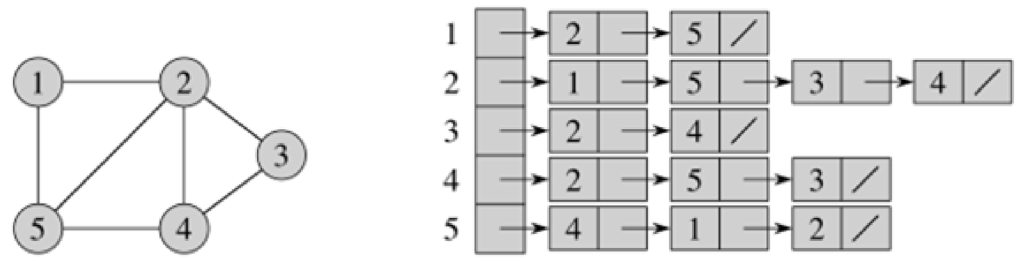
- A collection of benchmark datasets for the evaluation of graph kernels and graph neural networks

Freebase 15K (FB15k)

- A collection of 592,213 triplets with 14,951 entities and 1,345 relationships.

A simple representation for graphs

- A graph is often represented in computer as an adjacent matrix or adjacent list.

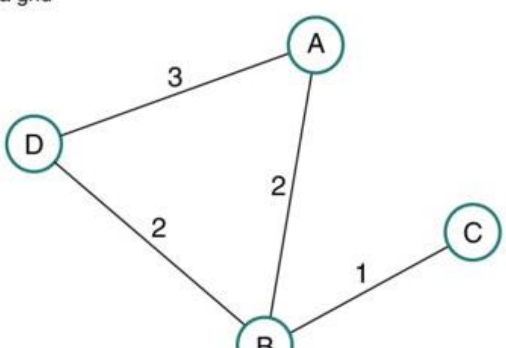


Rows	A	B	C	D
A	0	2	0	3
B	2	0	1	2
C	0	1	0	0
D	3	2	0	0

a) Square adjacency matrix as a grid

```
Rows,A,B,C,D
A,0,2,0,3
B,2,0,1,2
C,0,1,0,0
D,3,2,0,0
```

b) Matrix written to text file



c) Imported to Cytoscape Network

Output:

1: 2, 5

2: 1, 5, 3, 4

3: 2, 4

4: 2, 5, 3

5: 4, 1, 2

- Inappropriate for graphs containing complex relationships.
- Inefficient for computation and retrieval.

Graph databases: A definition

- A database that uses graph structures for semantic queries with nodes, edges, and properties to describe and store data
  - It is commonly referred to as a NoSQL database.

Graph DBs vs. Relational DBs

- Can graphs be stored in relational databases? No or difficult
- Then, why is it challenging?
- Important notes for relational databases
  - Each row has relations to other rows through primary key and foreign keys.
  - The data must conform a specified schema and a strict standard set of properties (ACID) $\rightarrow$ data is split into multiple tables.
  - Complex queries need joining several tables $\rightarrow$ tedious on large data
- Important notes for graph databases
  - Entities and relations need to be stored in simple representation, allowing fast accesses and queries.
  - The schema must be flexible to manage the data in an ad-hoc manner and adapt to a variety of scenarios where the data changes.
  - Easy to traverse on a series of edges

Graph DBs vs. Relational DBs

- Case study 1: Find the email addresses of users having phone numbers' region codes are 311.
- Relational databases
  - Find all the userIDs that have phone numbers' region codes are 311 in the Phones table.
  - For each userID found, find the matched EmailID in the User-Email table.
  - Then, look up the Emails table to get the email address for a given EmailID

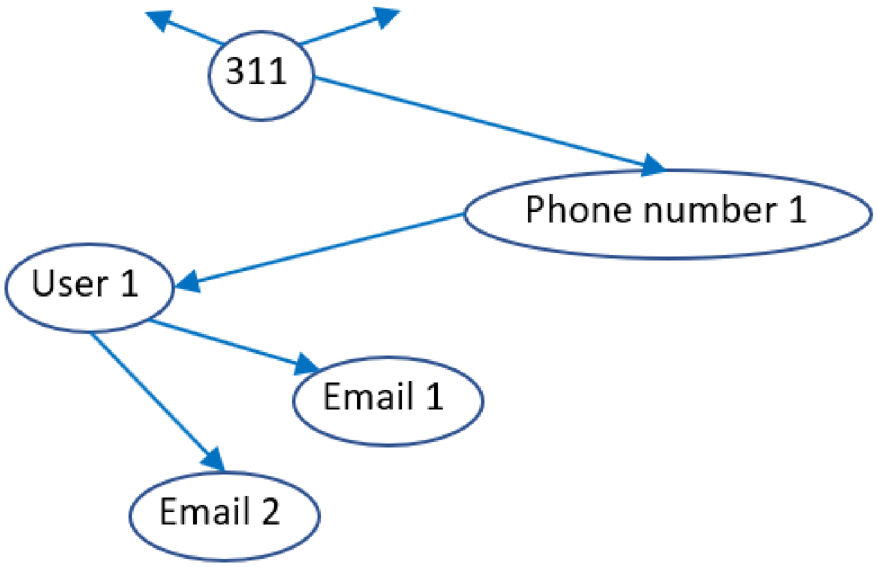
Phones	
UserID (PK)	Phone Number
...	...
...	...
...	...
...	...
...	...
...	...

User-Email	
UserID	EmailID
...	...
...	...
...	...
...	...
...	...
...	...

Emails	
EmailID	Email
...	...
...	...
...	...
...	...
...	...
...	...

Graph DBs vs. Relational DBs

- Case study 1: Find the email addresses of users having phone numbers' region codes are 311.
- Graph databases
  - Find the vertex that denotes the region code 311
  - Follow the backlink pointer to get the phone numbers
  - For each phone number found, follow the pointer to find the user
  - From the user, follow the email links to obtain the email addresses.



  - Only follow the link pointers, no join made $\rightarrow$ better computational saving

Graph DBs vs. Relational DBs

- Case study 2: List the names of all friends of the user “Jack”
- Relational databases: query by SQL statements
  - People (person_id, person_name) and Friend (friend_id, person_id)

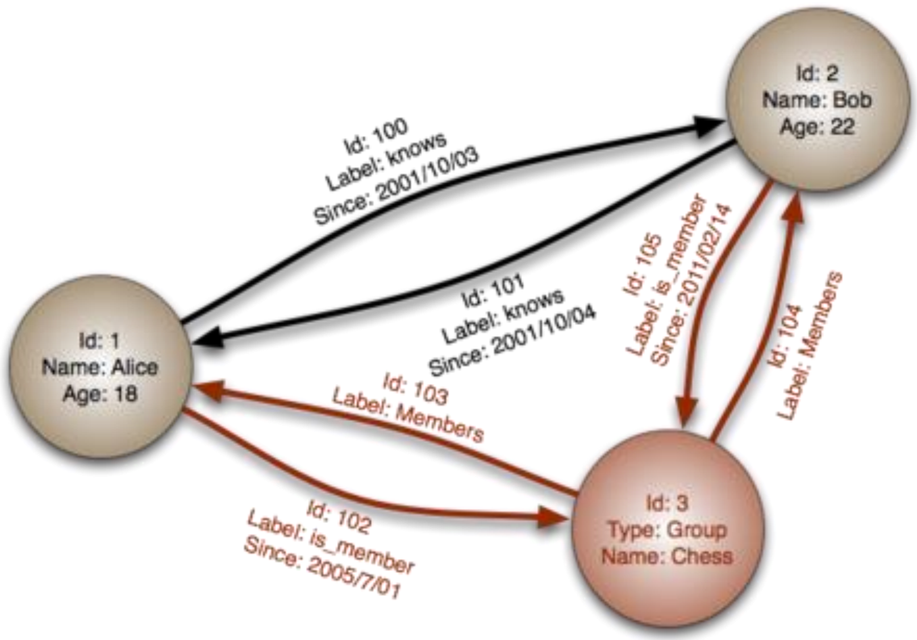
```
SELECT p2.person_name
FROM people p1
JOIN friend ON (p1.person_id = friend.person_id)
JOIN people p2 ON (p2.person_id = friend.friend_id)
WHERE p1.person_name = 'Jack';
```

- Graph databases: query by a designated query language (e.g., Cypher)

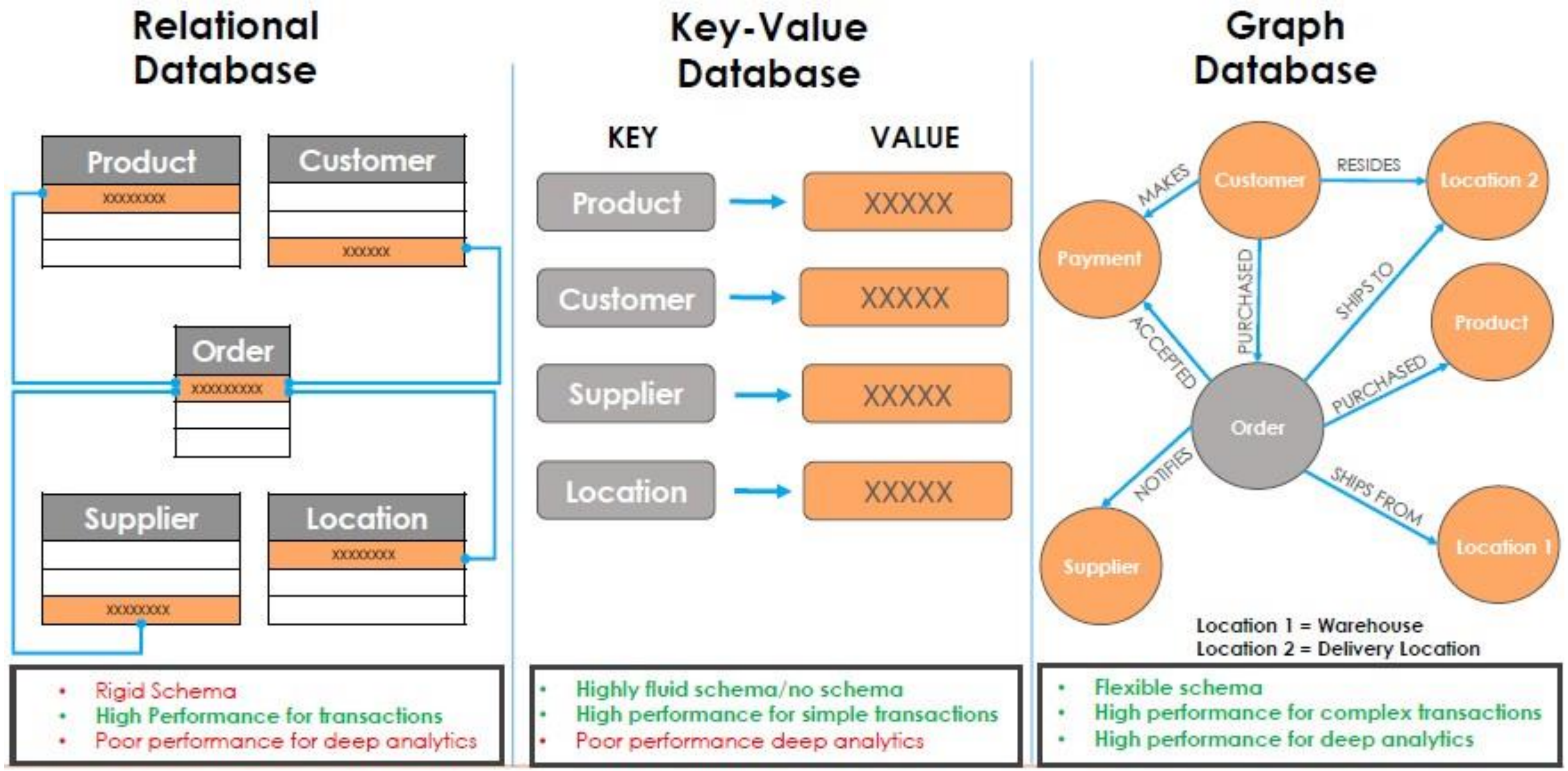
```
MATCH (p1:person)-[:FRIEND-WITH]-(p2:person)
WHERE p1.name = "Jack"
RETURN p2.name
```

Graph DBs vs. Relational DBs

- Relational databases are for flat data layout.
  - The relations in data are only 1-2 levels depth.
- Graph databases allow relations to be different depth levels.
  - Good for large-scale data and complex analysis task (e.g., OLAP)



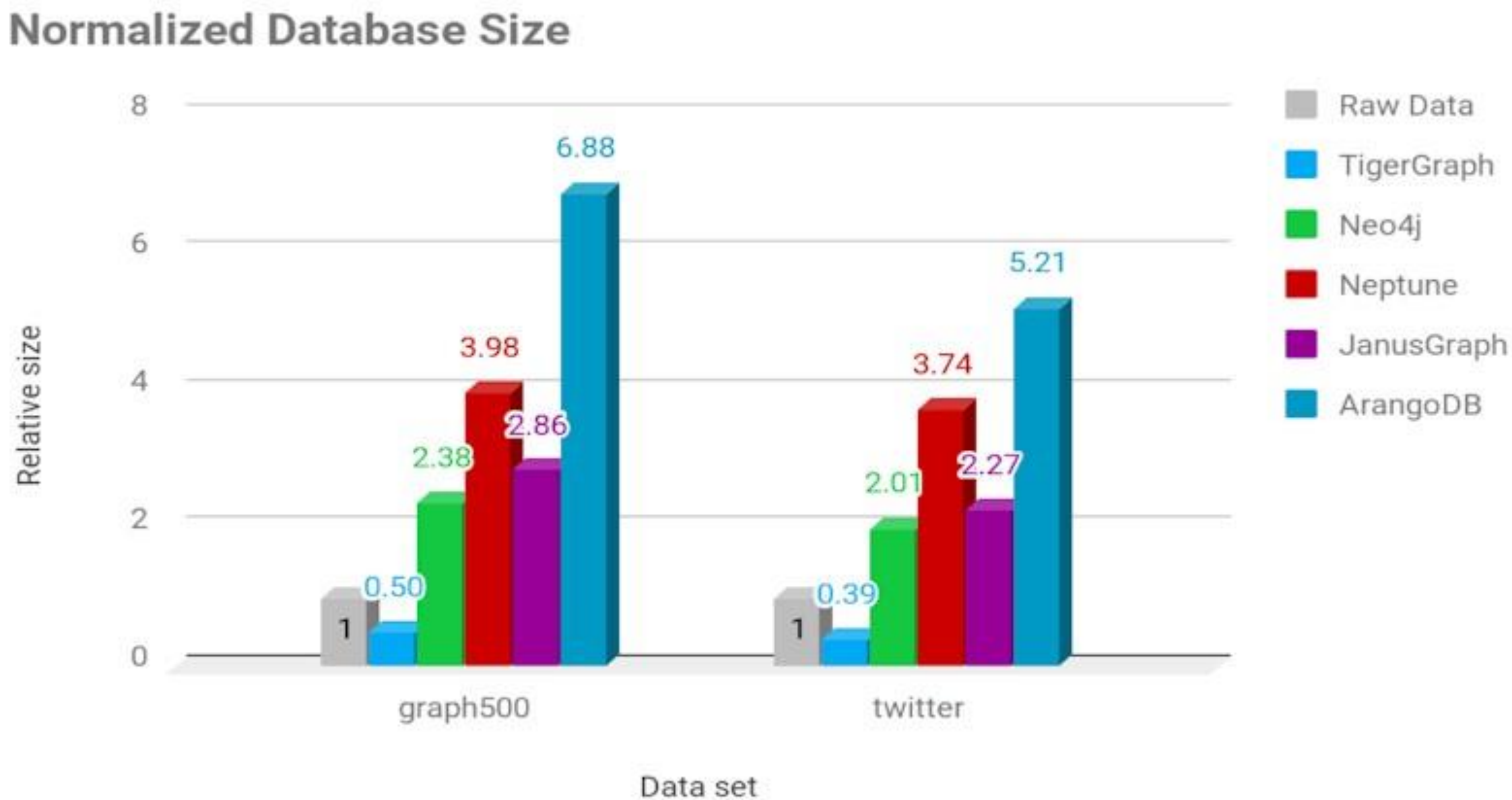
Graph database: Comparison



Graph databases: Evolution

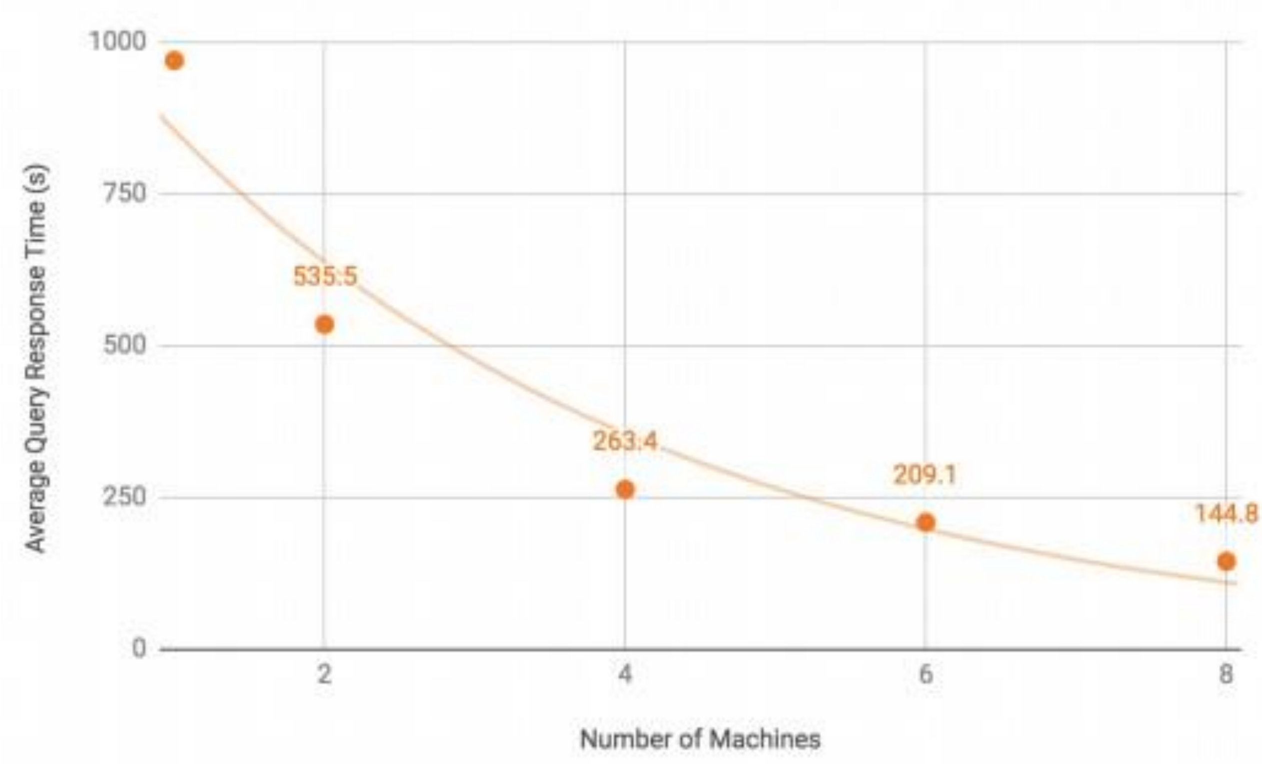
Evolution of Graph Databases	Graph 1.0 (e.g. Neo4j)	Graph 2.0 (e.g. Amazon Neptune, DataStax)	Graph 3.0 TigerGraph
Native Graph Storage	✓	✗ Built on key-value store	✓
Parallel Loading	✗ Days to load	✗ Days to load	✓ Hours to load terabytes
Parallel Multi-Hop Analytics	✗ Times out after 3+ hops	✗ Runs out of time/memory after 3+ hops	✓ Sub-second across 3-10+ hops
Parallel Updates (in real-time)	✗ Batch updates	✗ Batch updates	✓ Mutable/ Transactional
Scale up to Support Query Volume	✗ Queries can't go across clusters	✓ Scales for simple queries (1-2 hops)	✓ 2 billion+ per day in production
Privacy for Sensitive Data	✗	✗	✓ MultiGraph service

Graph database: Normalized sizes



A comparison of graph databases according to the sizes required for representing the original data

Graph database: TigerGraph



The response time of TigerGraph database w.r.t. the system scales

Graph tools: Spark GraphX and Plotly

GraphX d graph-parallel computation

- A collection of algorithms and builders to simplify graph analytic tasks
- Available only in Scala and RDD
- GraphFrames package extends GraphX to Python and DataFrames

- Alternatives: Apache Giraph, Gelly (Apache Flink)

plotly

- A tool for graphing, analytics, and statistics
  - Available for Python, R, MATLAB, Perl, Julia, Arduino, and REST
  - Plotly for Python is free and under the MIT license.

Graph tools: NetworkX and scikit-network

- A free Python library for studying graphs
  - Create, manipulate, and study of the structure, dynamics, and functions of complex networks

- A free Python libary for the analysis of large graphs
  - Memory-efficient representation as sparse matrices in the CSR format of scipy
  - Fast algorithms, simple API inspired by scikit-learn


...the end.